

Chiral *N*-*tert*-Butylsulfinyl Imines: New Discoveries

Francisco Foubelo^[a, b] and Miguel Yus^{*[b]}

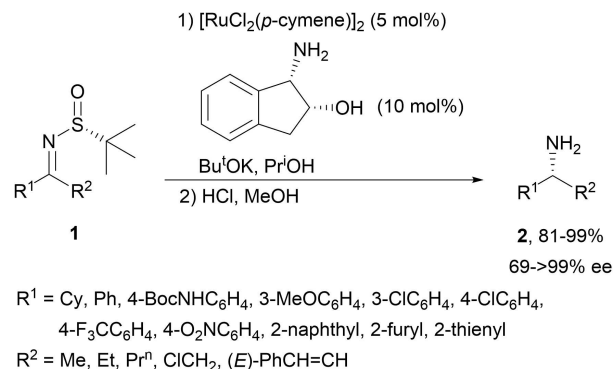
Abstract: In this account the reactions of chiral *N*-*tert*-butylsulfinyl imines with organometallic reagents such as organoalkaline (lithium, sodium, potassium and cesium derivatives), organomagnesium, organozinc, organoboron, organoaluminium, organoindium and organosilicon compounds is comprehensively described. The reactivity in all cases is derived to synthetic applications in order to prepare interesting organic nitrogenated molecules, especially in the field of alkaloid compounds.

Keywords: *N*-*tert*-butylsulfinyl imines, organoalkaline compounds, Grignard reagents, organozinc reagents, organoboron compounds, organoaluminum derivatives, organoindium and organosilicon compounds.

1. Introduction

Chiral *N*-*tert*-butylsulfinyl imines are interesting starting materials to prepare chiral amine derivatives by addition of different nucleophiles and final desulfinylation.^[1] These compounds are easily prepared by a Ti-catalyzed addition of commercially available chiral (*R*)- and (*S*)-*tert*-butylsulfinamide to carbonyl compounds under thermal^[2] or microwave^[3] conditions. On the other hand, the final desulfinylation can be easily carried out just by treatment with hydrochloric acid in an organic solvent (MeOH, ether, dioxane). In this chemistry the sulfinyl group acts both as an activating group and also as a chiral induction one. As nucleophiles several organometallic reagents derived from alkaline metals, organomagnesium and organozinc reagents, organoboron, aluminum, indium and silicon derivatives have been successfully used. This technology allows the asymmetric addition of nucleophiles to the C=N bond, so initially diastereomeric products are obtained in an enriched form. After desulfinylation, the corresponding chiral

primary amines are generated in an enantioenriched form. Another interesting aspect comes from the fact that both enantiomeric starting *N*-*tert*-butylsulfinyl imines are easily accessible from commercially available *tert*-butylsulfinamides, so allowing to prepare easily both series of products. This methodology has been widely used in the preparation of many chiral nitrogenated naturally-occurring compounds, such as alkaloids and amino acids. In this account, literature in the last decade on the use of *N*-*tert*-butylsulfinyl imines in asymmetric reactions with different nucleophiles to give chiral nitrogenated derivatives, which by desulfinylation are transformed into the corresponding chiral nitrogen-containing compounds, is re-



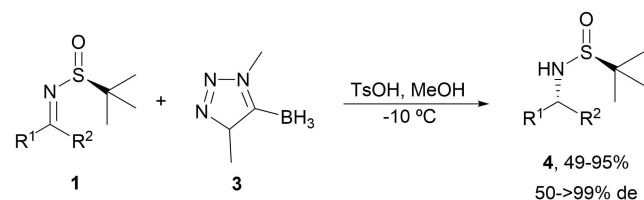
Scheme 1. Reduction of ketimines **1** to amines **2** by hydrogen transfer.

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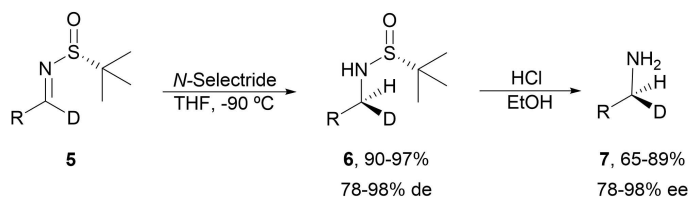
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R¹ = Ph, 4-MeC₆H₄, 4-MeOC₆H₄, 4-ClC₆H₄, 4-F₃CC₆H₄, 4-NCC₆H₄, 3-FC₆H₄,
2-HOC₆H₄, 2-MeOC₆H₄, 2-Py, 2-furyl, Bu^t, Ph(CH₂)₂, (*E*)-PhCH=CH
R² = Me, Et, Bu^t, Ph, ClCH₂

Scheme 2. Reaction of ketimines **1** with borane **3**.



R = Ph, 2-ClC₆H₄, 2-BrC₆H₄, 2-MeC₆H₄, 3-MeC₆H₄, 4-ClC₆H₄, 4-MeOC₆H₄,
2-furyl, 2-thienyl, *n*-C₆H₁₃CC, *n*-C₈H₁₇, Ph(CH₂)₂, Cy

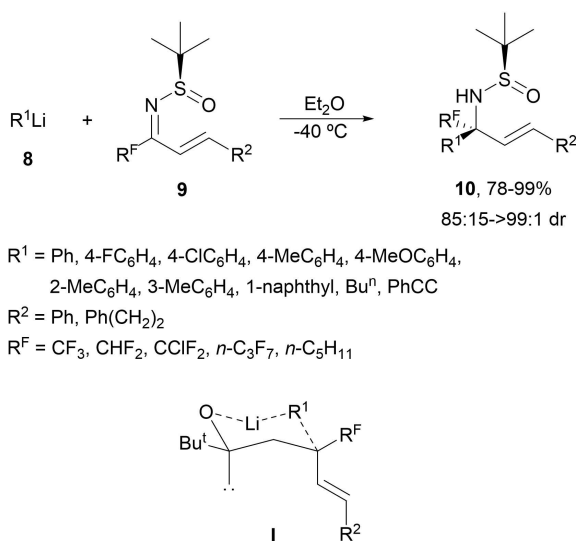
Scheme 3. Reaction of deuterated imines **5** with *N*-Selectride.



Francisco Foubelo was born in 1961. He studied chemistry at the University of Oviedo from which he received B.S. (1984), M.S. (1986), and Ph.D. (1989) degrees. After a postdoctoral stay (1989–1991) as a Fulbright fellow at Princeton University, he moved to the University of Alicante where he became Associate Professor in 1995 and Full Professor in 2002. Dr. Foubelo has co-authored more than 150 papers, and his current research interests are focused on the development of new synthetic methodologies involving chiral sulfinimines and on metal-promoted functionalization of alkenes and alkynes.



Professor Miguel Yus was born in Zaragoza (Spain) in 1947, and received his BSc (1969), MSc (1971) and PhD (1973) degrees from the University of Zaragoza. After spending two years as a postdoctoral fellow at the Max Planck Institut für Kohlenforschung in Mülheim a.d. Ruhr he returned to Spain to the University of Oviedo where he became associate professor in 1977, being promoted to full professor in 1987 at the same university. In 1988 he moved to a chair in Organic Chemistry at the University of Alicante. Professor Yus has been visiting professor at different institutions and universities among them ETH-Zentrum, Oxford, Harvard, Uppsala, Marseille, Tucson, Okayama, Paris, Strasbourg, Bologna, Sassari, Tokyo and Kyoto. He is co-author of around 650 papers (and six patents) and has supervised 62 Doctoral Theses, and delivered about 250 lectures, most of them abroad. His bibliometric data are more than 28.000 citations and h-index 77. He has received several international awards being also named Active Academician from the European Academy of Sciences and Arts, and Academic Member of the Athens Institute for Education and Research. Professor Yus has been in the Advisory Board of more than 30 international journals having been also Editor-in-Chief of Letters in Organic Chemistry and Open Chemistry. Professor Yus founded the new chemical company MEDAL-CHEMY S.L. to commercialize fine chemicals.



Scheme 4. Reaction of organolithium reagents **8** with fluorinated imine **9**.

ported. By this way, we would like to underline in this account the ability of this methodology to prepare chiral nitrogen-containing compounds which are in many cases of biological importance. Transition metal-catalyzed reactions, as well as other methodologies involving 1,3-dicarbonyl compounds, nitro compounds, radical reactions, or organocatalysis, will be not considered due to space limitations.

2. Reductions

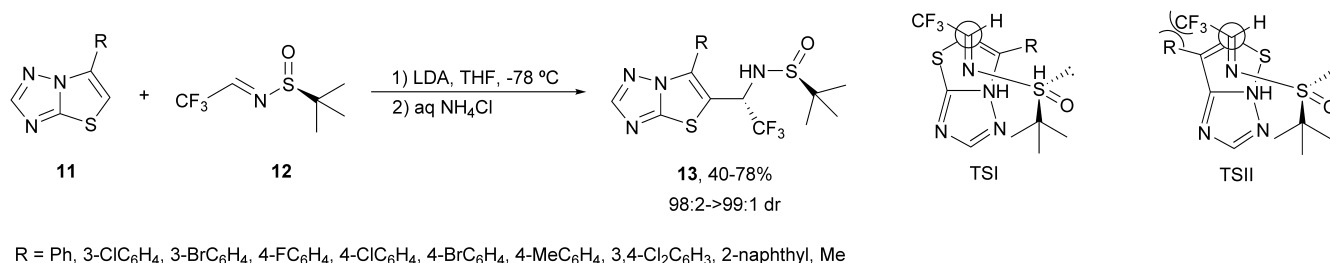
The hydrogen transfer to ketimines **1**^[4,5] using $[\text{RuCl}_2(p\text{-cymene})]_2$ as catalyst, an aminoindanol as ligand, isopropanol as hydrogen source and Bu^tOK as base led to the formation of the corresponding primary amines **2** after desulfinylation (Scheme 1).^[6] Starting from *ent*-**1** the corresponding amines *ent*-**2** were obtained with similar results.

An interesting simplification of the reaction shown in Scheme 1 consists in the use of an achiral amino alcohol, namely 2-amino-2-methylpropanal, as ligand in the same reaction, to yield the corresponding amines **2**, with similar results (71–99 %, 65–>99 % ee).^[7] DFT calculations indicates that the process shown in Scheme 1 is not concerted: it occurs through two transition states and three intermediates.^[8] On the other hand, a microwave version of the reaction has been already reported.^[9]

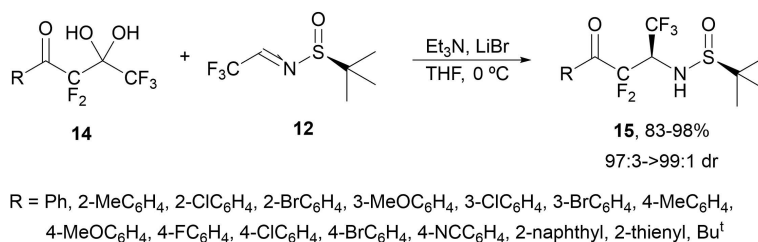
The chemistry depicted in Scheme 1 has been applied to the synthesis of lactams starting from *N*-*tert*-butylsulfinyl iminoesters,^[10] or different nitrogenated heterocycles, from *N*-*tert*-butylsulfinyl haloimines.^[11]

The asymmetric reduction of ketimines **1** with *N*-heterocyclic carbene boranes **3** afforded compounds **4** in good yields and stereoselectivities (Scheme 2).^[12] After deprotection, the corresponding primary amines **2** were obtained (84–90 %).

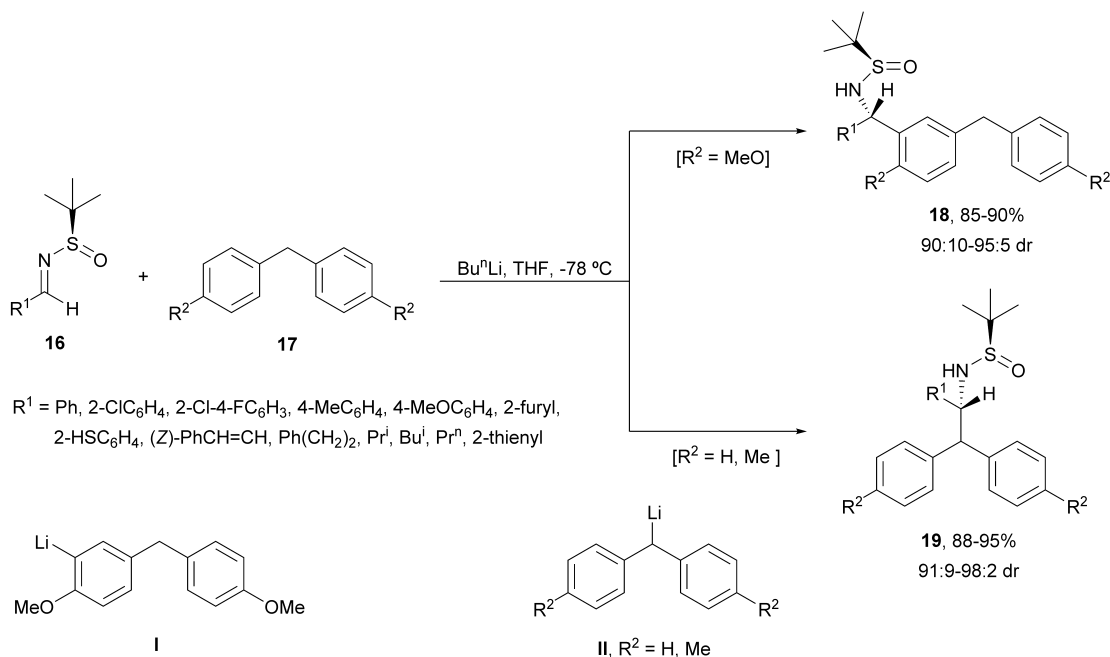
Optically active deuterated primary amines **7** were easily prepared starting from deuterated *N*-*tert*-butylsulfinyl aldimines **5**, by reduction with *N*-selectride, to give **6**, followed by desulfinylation with HCl in EtOH (Scheme 3).^[13]



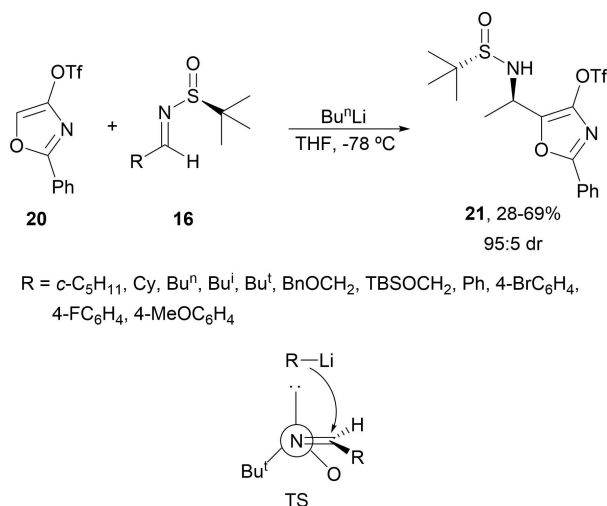
Scheme 5. Enantioselective reaction of heterocycles **11** with imine **12** in the presence of LDA.



Scheme 6. Reaction of compounds **14** with imines **12**.



Scheme 7. Reaction of imines **16** with diarylmethanes **17**.



Scheme 8. Reaction of isoxazoles **20** with imines **16**.

From the results included in this Section we conclude that the hydrogen transfer to Ellman ketimines is a useful procedure to produce, after final desulfinylation, primary amines in an enantioselective enriched form. In addition boranes or *N*-Selectride also reduced imines to give the corresponding chiral primary amines.

3. Organoalkaline Compounds

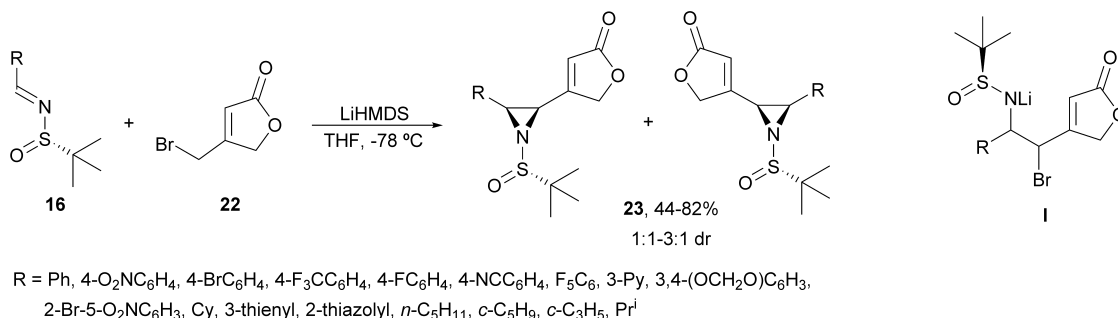
3.1. Organolithium Compounds

Aliphatic and aromatic organolithium compounds **8** react with fluoroalkyl unsaturated imines **9** to give mainly products **10** with high diastereoselectivities (Scheme 4).^[14] Transition state **I** can explain the obtained results. In addition, compounds **10** are easily transformed into the corresponding amino acids by a sequence of desulfinylation/oxidation of the C=C bond.

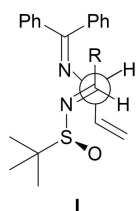
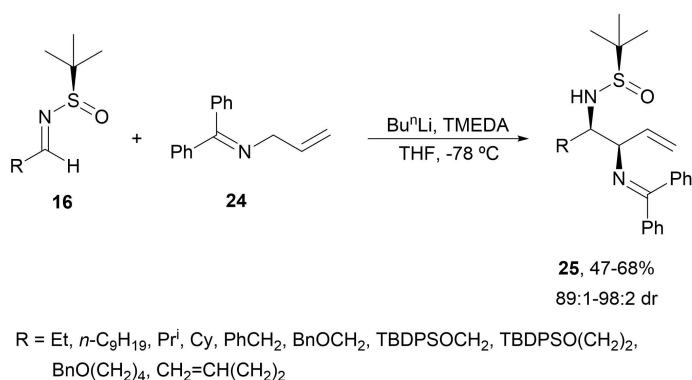
An asymmetric addition of the anion derived from substituted thiazolotriazoles **11** to trifluoromethyl imines **12** led to the formation of compounds **13** in excellent diastereoselectivities (Scheme 5).^[15] A non-chelated transition state TSI is preferred compared to TSII, due to steric interaction of groups R and CF₃.

The same imines **12** react with fluorinated materials **14** to yield the expected products **15** with excellent yields and diastereoselectivities (Scheme 6).^[16] The transition state TS depicted in Scheme 6 can explain the stereochemistry observed in compounds **15**.

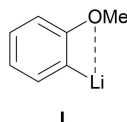
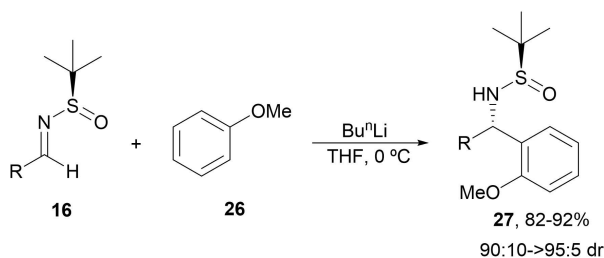
The reaction of sulfinyl imines **16** with diarylmethanes **17** in the presence of BuⁿLi gave different products depending on the substitution of the aromatic rings: for methoxy substituents, the *ortho*-reaction took place giving products **18**, whereas for non-substitution, or methyl substituents, the reaction occurred at the methylenic group, yielding compounds **19** (Scheme 7).^[17] Intermediates **I** and **II**, respectively, are responsible of the obtained products.



Scheme 9. Reaction of imines **16** with bromomethylbutanolid **22**.



Scheme 10. Reaction of imines **16** with allylimine **24**.



Scheme 11. Reaction of lithiated anisole with imines **16**.

A similar methodology was applied to oxazolyl triflates **20**, in the reaction with *N*-sulfinyl imines **16**, so the corresponding heterocycles **21** were obtained with high stereoselectivities (Scheme 8).^[18] An open-chelate transition state TS was proposed to explain the obtained results.

The asymmetric vinilogenous^[19] aza-Darzens reaction between imines **16** and bromomethylbutanolid **22** occurred in the presence of LiHMDS yielding a mixture of *cis*-aziridines **23** (Scheme 9).^[20] As intermediates, the corresponding salts **I** were proposed to take part in the process.

The reaction of imines **16** with allylimine **24** in the presence of BuⁿLi and TMEDA led to the formation of compounds **25** (Scheme 10).^[21] In the case of R=Br(CH₂)₄, the reaction gave a piperidine derivative, which after some transformation, led to (–)-epiquinamide. The stereochemistry of the obtained products was explained through the model **I**.

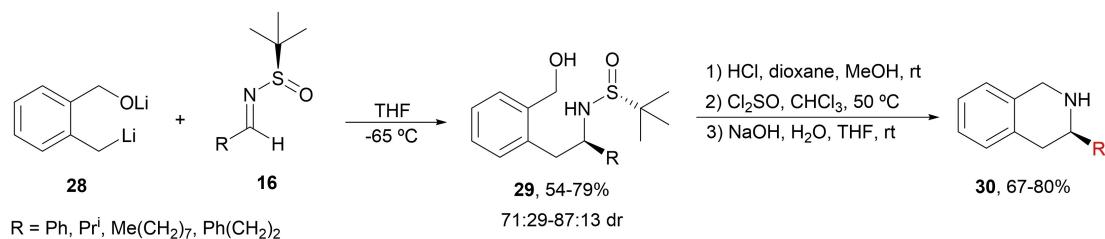
When the starting sulfinyl imine contains a α-DTBSO group, the reaction with different lithiated heterocycles led to the formation of protected chiral 1,2-amino alcohols.^[22] In the case of anisole (**26**), after lithiation with BuⁿLi, the reaction with imines **16** led to the formation of the expected compounds **27** with high stereoselectivities (Scheme 11).^[23] Intermediate **I** seems to be involved in the process.

Finally, a functionalized organolithium compound^[24] **28** (prepared by a DTBB catalyzed^[25] ring opening of phthalan^[26]) reacted with an imine **16** to yield compounds **29**, which by desulfinylation and cyclization, afforded tetrahydroisoquinolines **30** (Scheme 12).^[27]

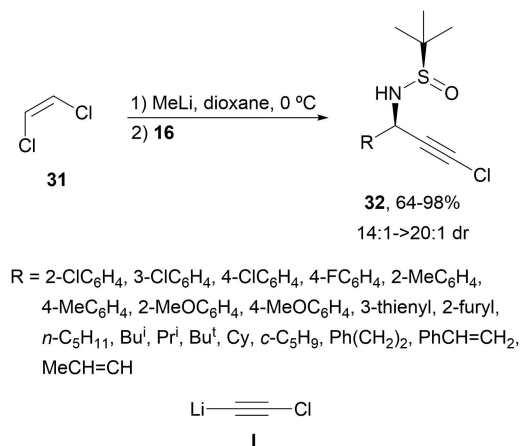
3.2. Alkynyllithiums

In situ generated lithium chloroacetylide **I** from (Z)-1,2-dichloroethylene **31** reacted with imines **16**, to give chlorinated propargyl amines **32** with high stereoselectivities (Scheme 13).^[28]

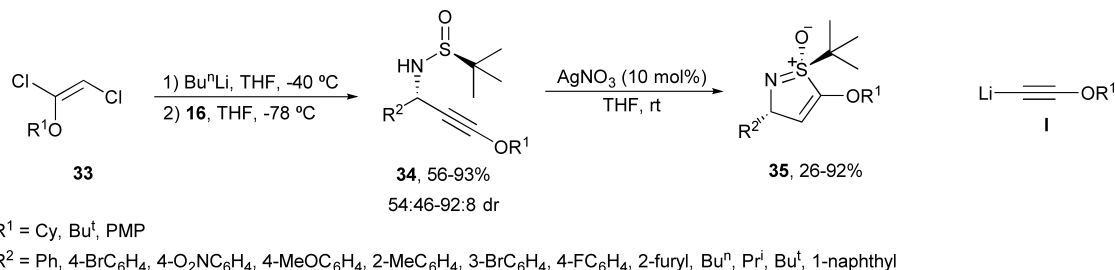
When the starting material contains an alkoxy group, like in **33**, the same reaction with imines **16** led to the formation of the corresponding products **34** (Scheme 14).^[29] Intermedi-



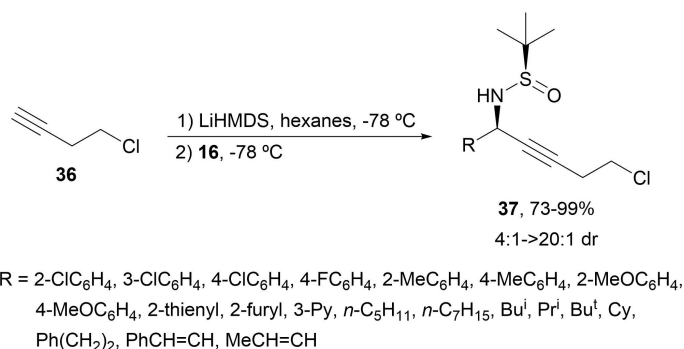
Scheme 12. Reaction of organolithium compound **28** with imines **16** and further cyclization.



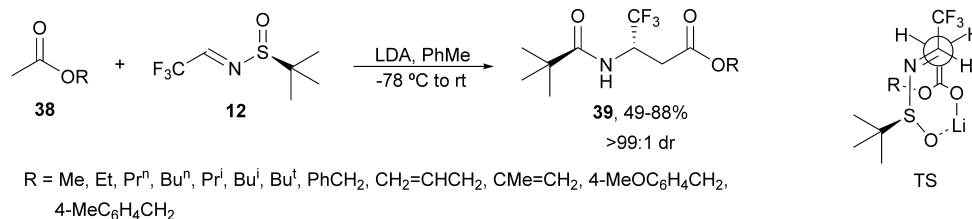
Scheme 13. Reaction of dichloroethylene **31** with imines **16** in the presence of MeLi.



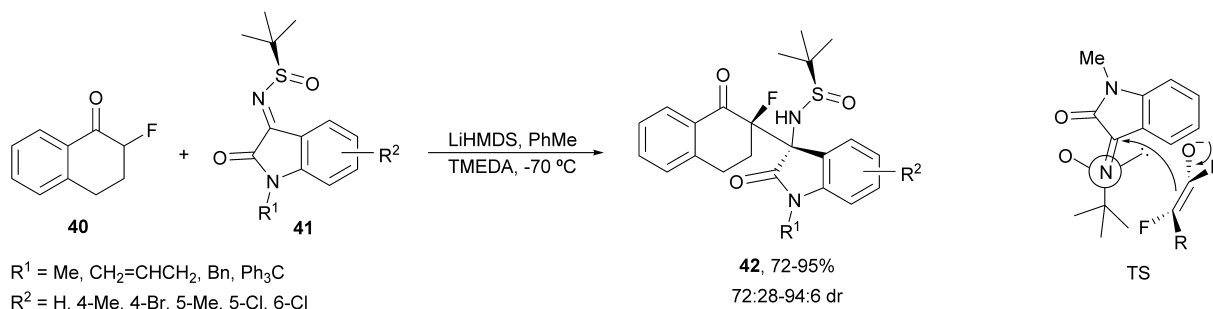
Scheme 14. Reaction of intermediates **I** with imines **16**.



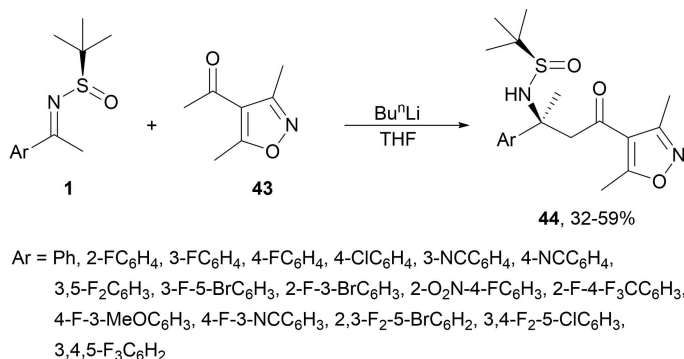
Scheme 15. Reaction of chloroalkyne **36** with imines **16**.



Scheme 16. Reaction of enolate derived from **38** and imine **12**.



Scheme 17. Reaction of enolate derived from **40** with imines **41**.

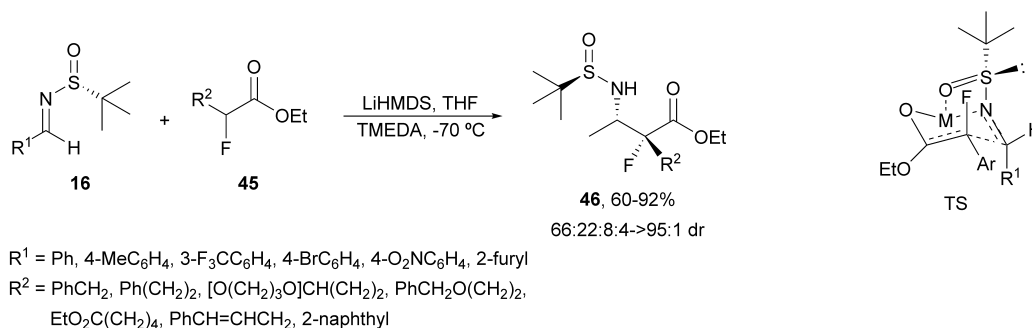


Scheme 18. Reaction of enolate derived from **43** with ketimines **1**.

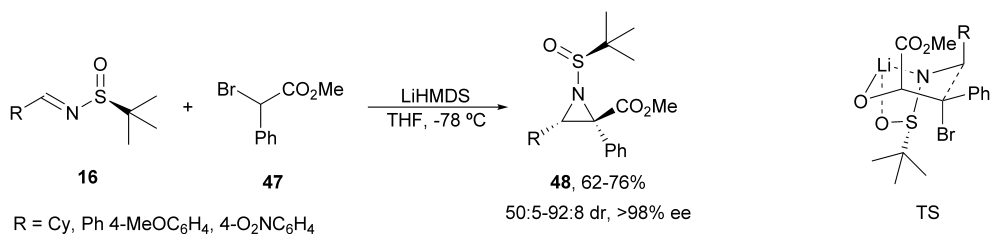
hydrolysis, the corresponding products **39** with excellent diastereoselectivities (Scheme 16).^[33] The reaction was also extended to α,α -disubstituted esters with similar results (44–99 %, 90 : 10–>99:1 dr). Transition state TS can explain the observed stereochemical outcome.

A diastereoselective Mannich reaction of α -fluoro ketones **40** with ketimines **41** afforded products **42** (Scheme 17).^[34] In order to explain the obtained stereochemistry an open transition state TS has been proposed.

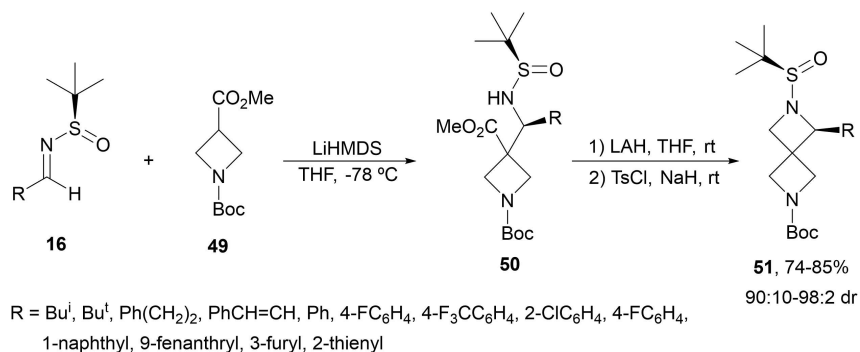
Another lithium enolates derived from oxazoles **43** were reacted with ketimines **1** to yield the expected products **44** (Scheme 18).^[35] The reaction was also extended to other



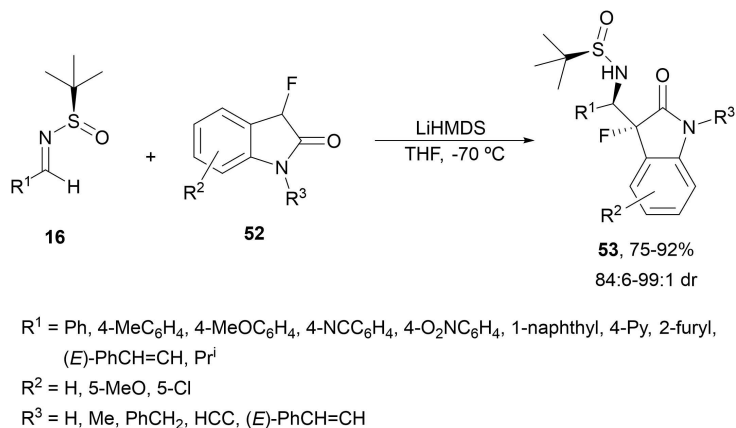
Scheme 19. Reaction of enolate derived from fluoro esters **45** with imines **16**.



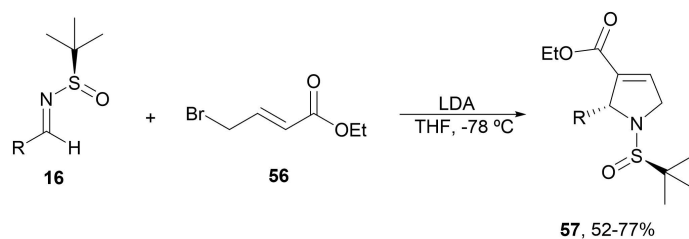
Scheme 20. Reaction of imine **16** with α -bromo ester **47**.



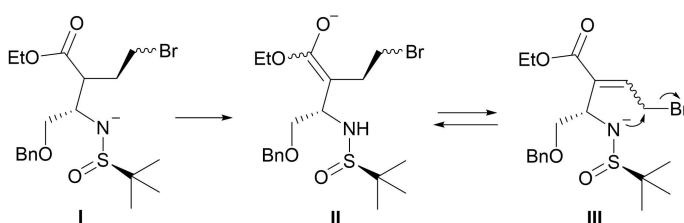
Scheme 21. One-pot reaction of imines **16** with ester **49**, followed by cyclization of compounds **50** to spiro products **51**.



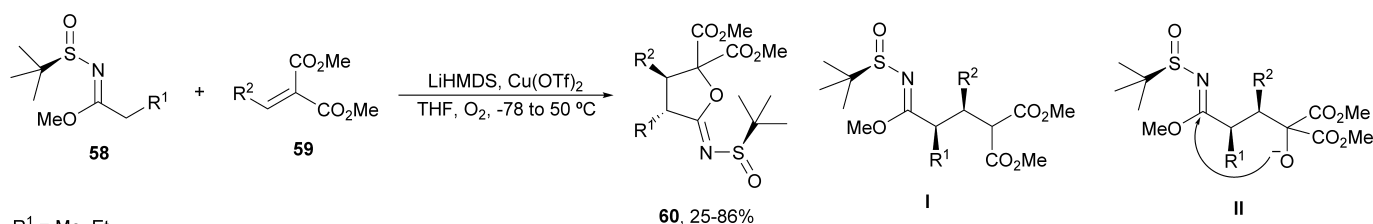
Scheme 22. Reaction of imines **16** with fluoro compounds **52** or **54**.



R = Ph, BnO, TBSO(CH₂)₃, *n*-C₆H₁₃, Prⁱ, Ph(CH₂)₂, Me₂C=CH(CH₂)₂C(Me)CH₂



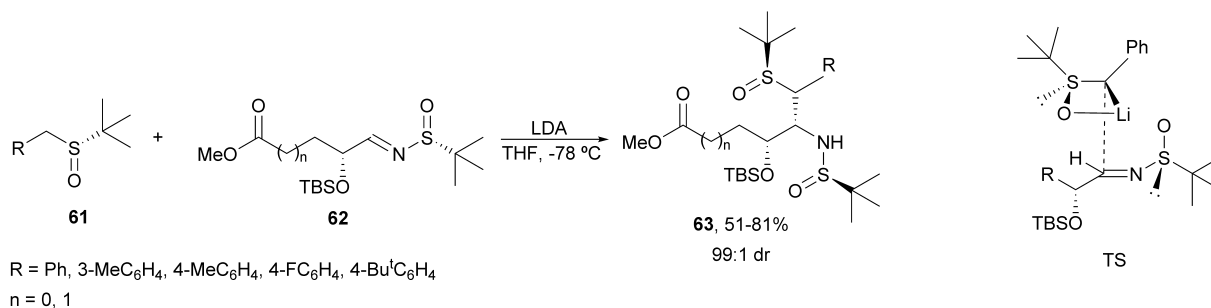
Scheme 23. Reaction of imines **16** and ethyl bromocrotonate **56**.



R¹ = Me, Et

R² = Ph, 2-FC₆H₄, 4-FC₆H₄, 4-MeC₆H₄, 2-MeOC₆H₄, 3-MeOC₆H₄, 4-MeOC₆H₄, 3,4-(MeO)₂C₆H₃, 1-naphthyl, 2-naphthyl, 2-thienyl, 2-furyl, 3-furyl

Scheme 24. Reaction of imidates **58** with unsaturated diesters **59** under oxidative conditions.



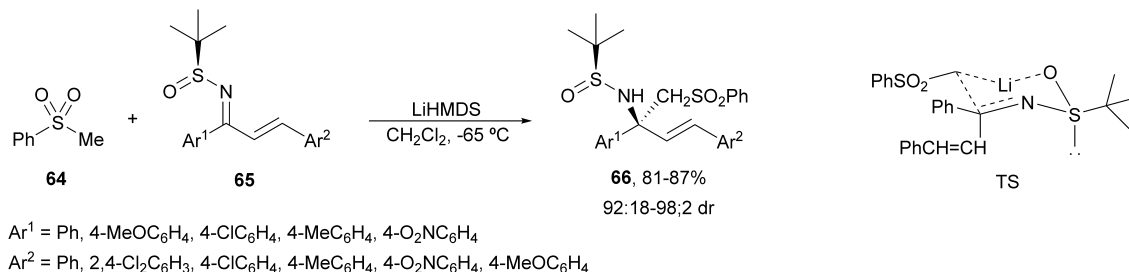
R = Ph, 3-MeC₆H₄, 4-MeC₆H₄, 4-FC₆H₄, 4-Bu^tC₆H₄
n = 0, 1

Scheme 25. Reaction of the enolate of **61** with imines **62**.

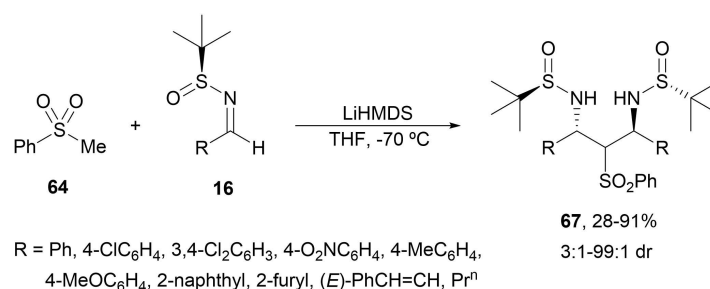
imines derived from heterocyclic compounds, as well as other substituted oxazoles with similar results.

A diastereoselective Mannich-type reaction occurred between α -fluoro esters **45** and imines **16** in the presence of LiHMDS and TMEDA, affording the expected products **46** with very good diastereoselectivities (Scheme 19).^[36] Curiously,

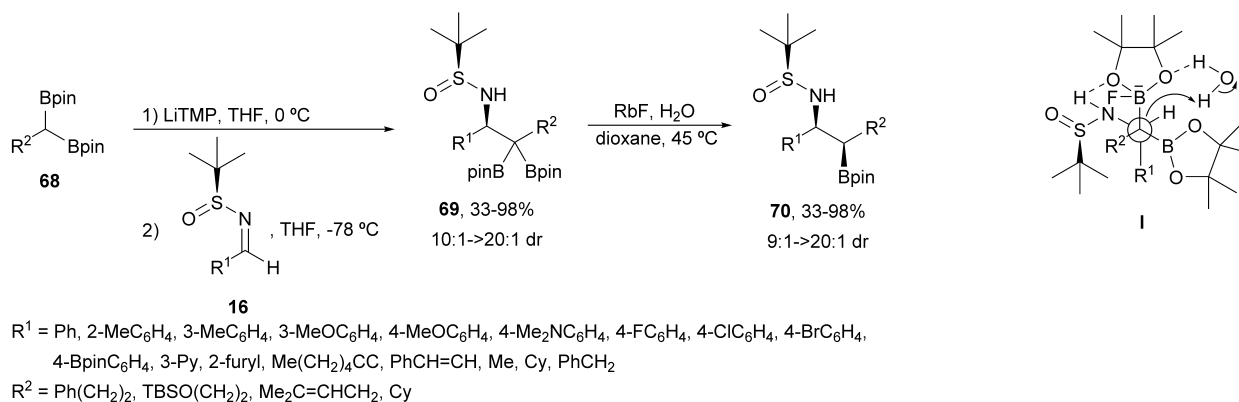
for aromatic substituted esters **45** (R=Ar), the stereochemistry of the C–N bond was the opposite than for aliphatic materials. Whereas an open transition state (Scheme 17, TS) can justify the obtained results for aliphatic systems, in the case of aromatic derivatives, a chair-like transition state has been proposed to explain the obtained results.



Scheme 26. Reaction of sulfone **64** with imines **65**.



Scheme 27. Reaction of sulfone **64** with imines **16**.



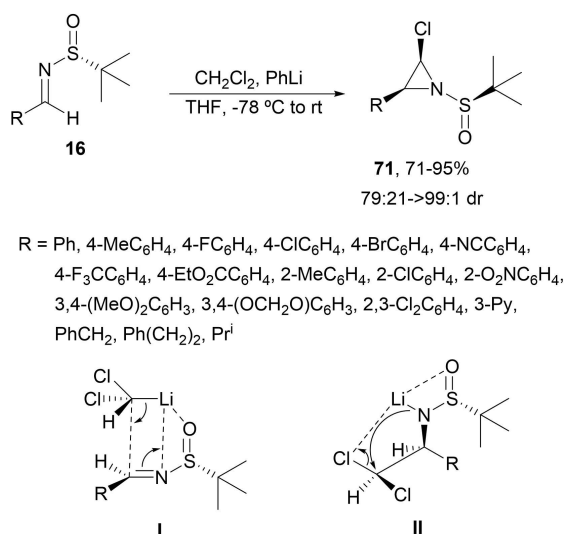
Scheme 28. Reaction of lithiated **68** with imines **16** followed by protodeboronation.

In the same way, fluorinated imines **12** reacted with enolates derived from malonates to yield the corresponding products with high diastereoselectivity.^[37]

The aza-Darzens reaction of imines **16** with α -bromo esters **47** in the presence of LiHMDS afforded mainly (*E*)-aziridines **48** with moderate-to-good diastereoselectivities (Scheme 20).^[38] For $\text{R}=\text{propylidene}$, the corresponding vinyl aziridines were obtained with similar results. As reported already,^[39] Scheme 20 shows the possible transition state TS which can explain the stereochemistry observed in the process.

The first step in the preparation of diazspiro[3.3]heptanes **51** is the reaction of imines **16** with azetidine **49** in the presence of LiHMDS at low temperature. After *in situ* treatment of compounds **50** with TsCl/NaH , the expected products **51** were obtained (Scheme 21).^[40]

Imines **16** react with the enolate derived from fluoroindoles **52** to give the corresponding products **53** with excellent diastereoselectivities (Scheme 22).^[41] The same process was also studied with fluoromethyl amide **54** to yield compounds **55** also with high diastereoselectivities. The stereochemistry observed could be explained through the



Scheme 29. Reaction of imines **16** with *in situ* generated Cl_2CHLi .

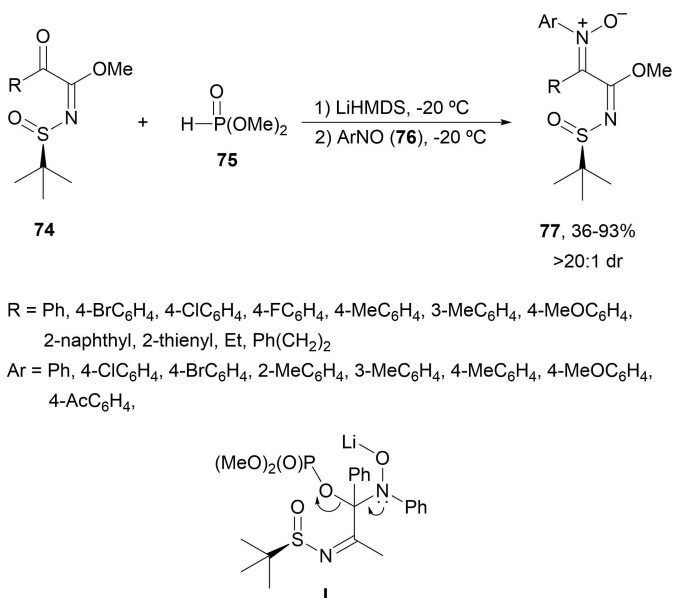
transition states shown in Schemes 17 and 19, for compounds **53** and **55**, respectively.

The reaction of imines **16** (or *ent*-**16**) with ethyl 4-bromocrotonate (**56**) led to the formation of pyrrolines **57** (or *ent*-**57**) through a [3+2] annulation process (Scheme 23).^[42] Intermediates **I–III** seem to take part in the reaction.

The enolate derived from imidate **58** reacted with unsaturated diesters **59** to give methyl substituted butyrolactonimides **60**, after an *in situ* oxidation in the presence of $\text{Cu}(\text{OTf})_2$ (Scheme 24).^[43] Intermediates **I** and **II** seem to take part in the process.

3.3.2. Sulfur Containing Compounds

Other lithiated carbanions derived from sulfoxides **61** were added to imines **62** to yield compounds **63** with excellent

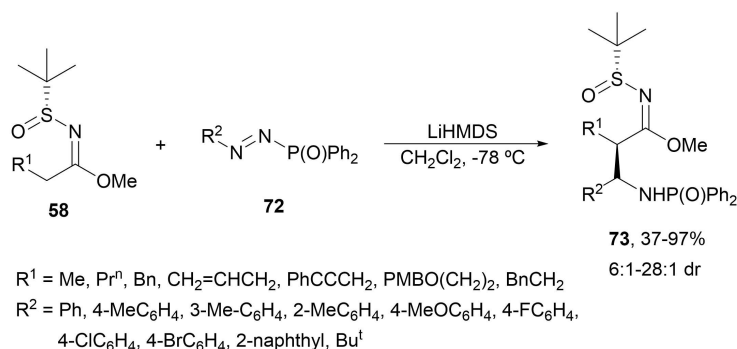


Scheme 31. Reaction of imidates **74** with phosphite **75** and LiHMDS followed by treatment with a nitroso arene **76**.

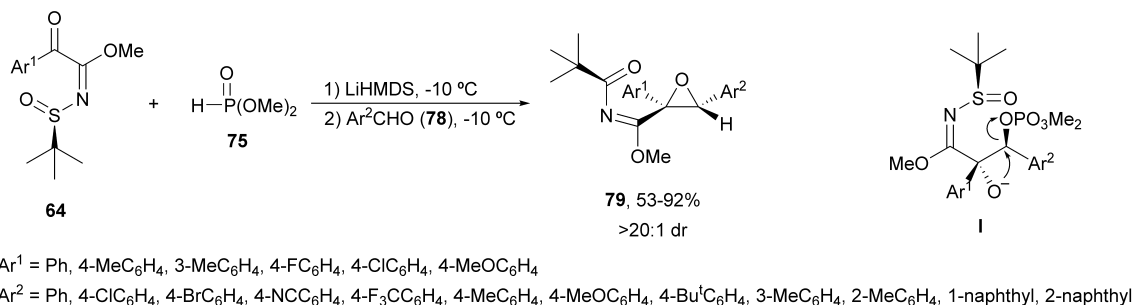
diastereoselectivities (Scheme 25).^[44] The reaction, which failed for 2-substituted aromatic sulfoxides **61**, seems to take place through the transition state depicted in Scheme 25, which offers less steric hindrance than in other topologies.

In the case of methyl phenyl sulfone (**64**), the corresponding α -lithio derivative reacts with different imines **65**, to yield the corresponding products **66**, with very good diastereoselectivities (Scheme 26).^[45] The six-membered transition state depicted in Scheme 26, with the Bu^t group occupying an equatorial position, is proposed to explain the observed stereochemistry. On the other hand, it is remarkable that no 1,4-addition was observed in any case.

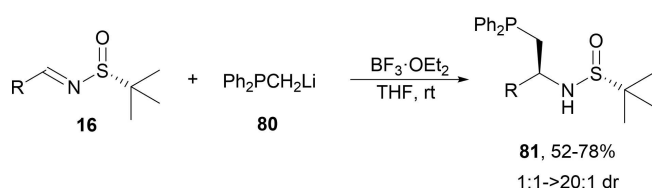
The same reaction shown in Scheme 26 was applied to imines **16**, so a double addition took place giving the



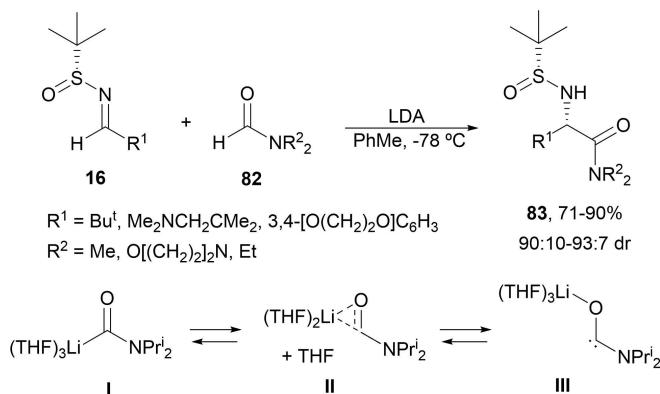
Scheme 30. Reaction of imidates **58** with azenes **72** in the presence of LiHMDS.



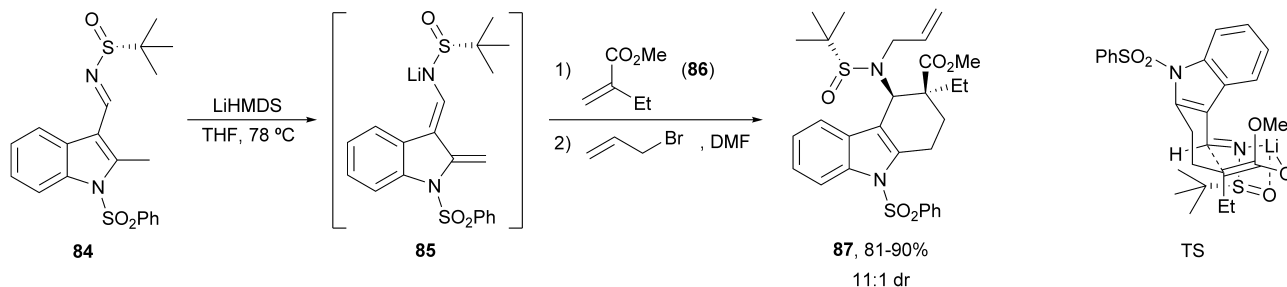
Scheme 32. Reaction of imidates **74** with phosphite **75** and aldehydes **78**.



Scheme 33. Reaction of imines **16** with Ph₂PCH₂Li (**80**).



Scheme 34. Reaction of lithium carbamoyl from formamides **82** with imines **16**.



Scheme 35. Reaction of deprotonated imine **85** with acrylate **86**.

corresponding sulfonylated diamines **67** (Scheme 27).^[46] Desulfonylation of compounds **67** under radical conditions (Buⁿ₃SnH/AIBN) gave sulfonylated diamines with C₂ symmetry (R=Ph, 4-MeC₆H₄; 60–63 %).

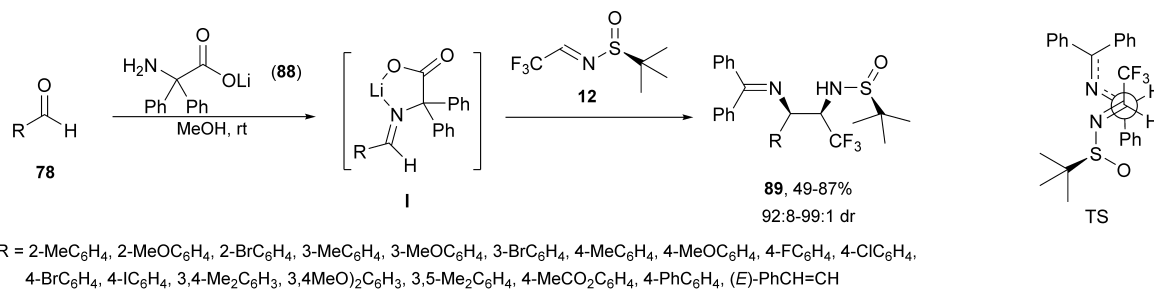
3.3.3. Other Lithium Intermediates

The reaction of lithiated 1,1-diborylalkanes **68** with imines **16** led to the formation of compounds **69**, which by deboronation yielded boryl amines **70** (Scheme 28).^[47] Based on ¹¹B-NMR experiments, a model **I** was proposed to explain the stereochemistry observed for compounds **70**.

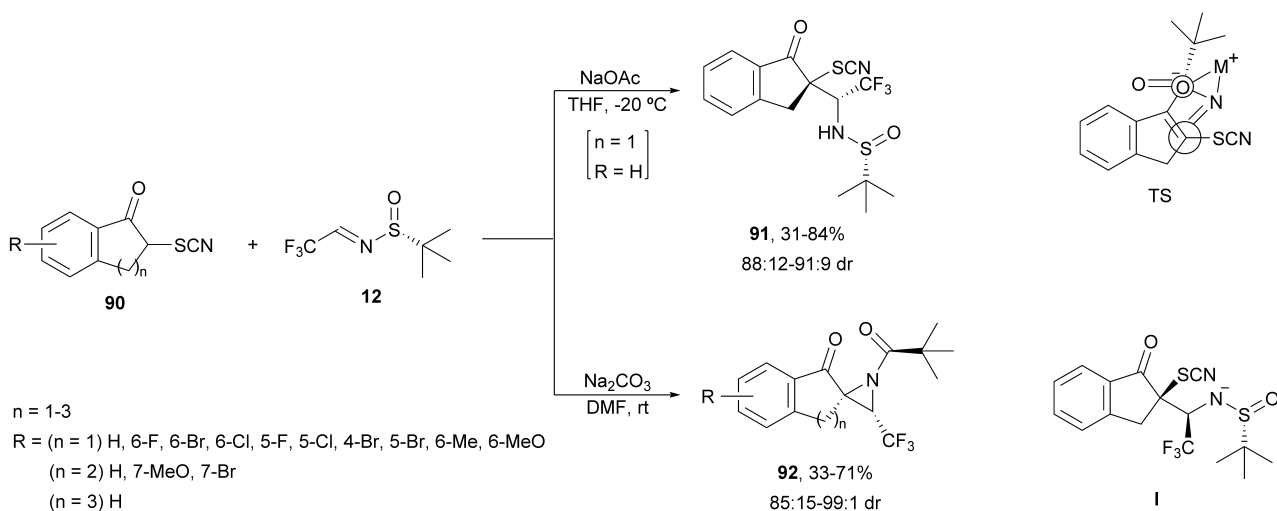
When imines **16** were allowed to react with *in situ* generated Cl₂CHLi (by reaction of CH₂Cl₂ and PhLi), the formation of aziridines **71** took place in excellent diastereoselectivities (Scheme 29).^[48] Transition states **I** and **II** have been proposed in order to explain the obtained results.

Imidates **58** react with diazenes **72** in the presence of LiHMDS to yield compounds **73** (Scheme 30).^[49] Similar results were obtained using NaHMDS/nitroso arene under the same reaction conditions.

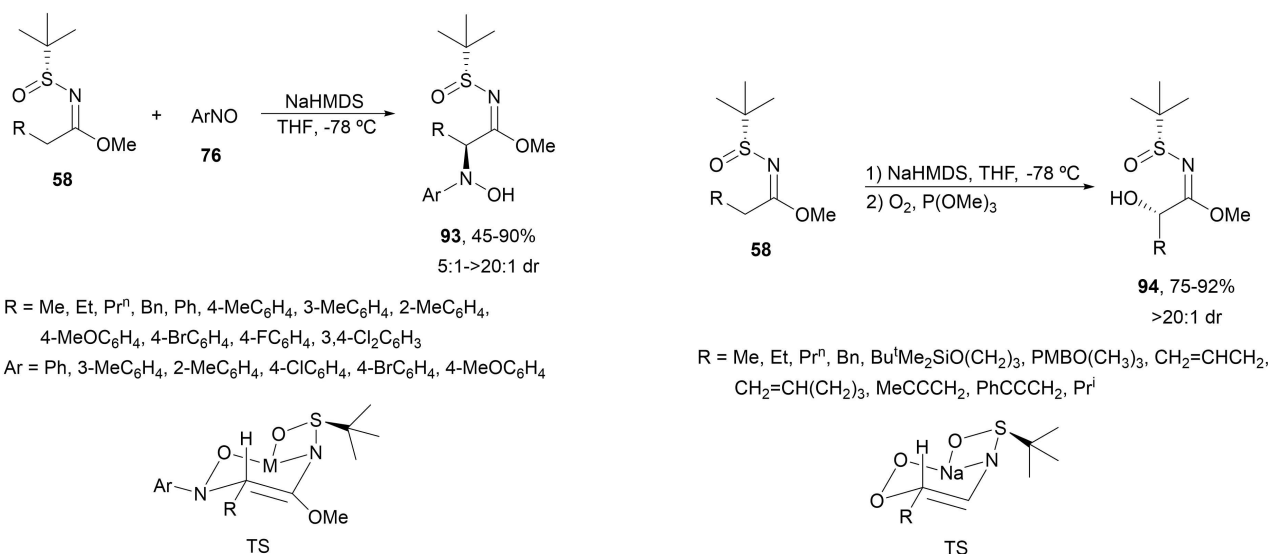
The treatment of keto imidates **74** with dimethyl phosphite (**75**) and LiHMDS, followed by reaction with an aromatic nitroso arene **76**, led to the formation of nitrones **77** with excellent diastereoselectivity (Scheme 31).^[50] Through ¹⁸O-labelling experiments, intermediate **I** has been proposed to explain the obtained results.



Scheme 36. Reaction of intermediates **I** with imine **12**.

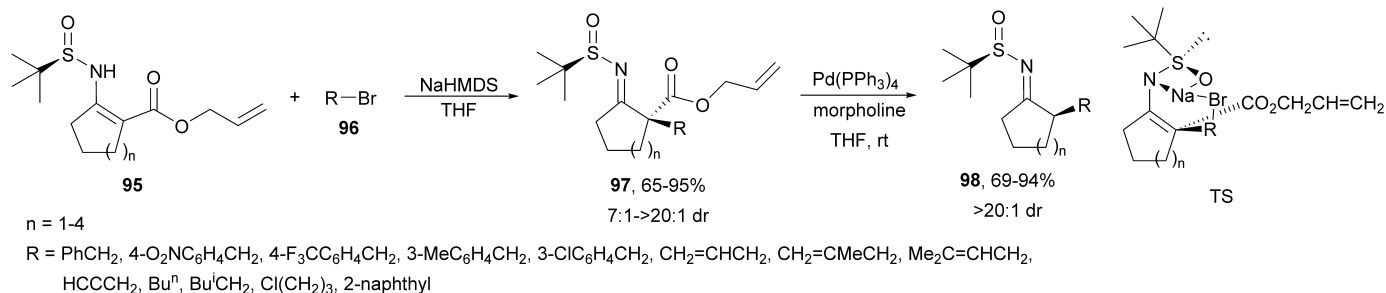


Scheme 37. Reaction of sodium enolates derived from **90** with imine **12**.

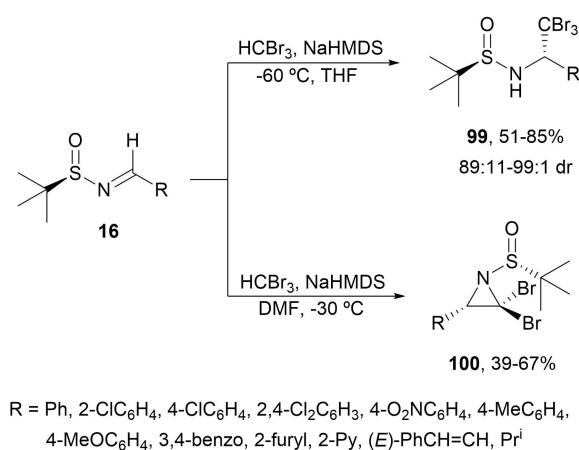


Scheme 38. Reaction of enolates derived from **58** with nitroso arenes **76**.

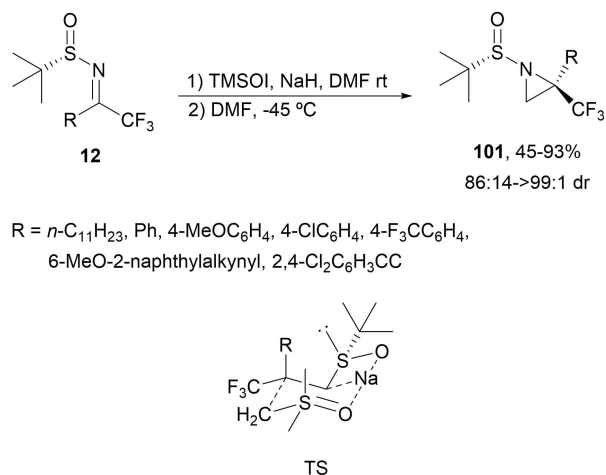
Scheme 39. Reaction of the enolate derived from **58** with oxygen.



Scheme 40. Reaction of sulfinyl enamines **95** with RBr (**96**) and further deprotection.



Scheme 41. Reaction of imines **16** with HCBBr₃ and NaHMDS.



Scheme 42. Reaction of imines **12** with TMSOI and NaH.

The same group has reported the reaction shown in Scheme 31, but using an aldehyde **78** instead of the nitroso compound, so epoxide **79** was diastereoselectively obtained (Scheme 32).^[51] Transition state TS has been proposed to explain the obtained results.

The addition of Ph₂PCH₂Li (**80**) to imines **16** yielded the corresponding products **81** with modest-to-excellent diastereoselectivities (Scheme 33).^[52] After protection of the phosphorous atom with BH₃, the desulfinylation led to the formation of 1,2-aminophosphines, interesting chiral ligands in transition-metal promoted reactions.

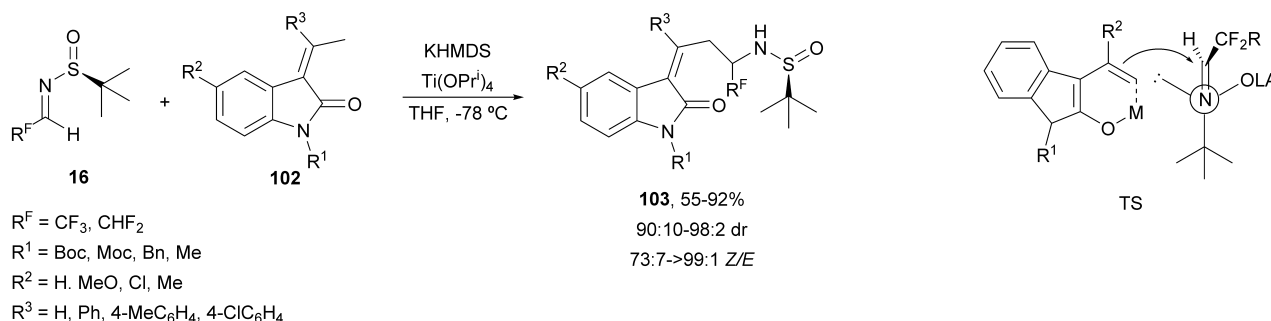
In situ generated carbamoyllithium (from amides **82**) was added to imines **16** to yield the corresponding products **83** (Scheme 34).^[53] The same process applied to ketimines afforded the expected products also with excellent yields (67–81 %) and diastereoselectivities (92:8–98:2 dr). Intermediates **I–III** have been postulated for the reaction depicted in Scheme 34.

Deprotonation of imino compound derived from 2-methyl indol **84** with LiHMDS yields the amino diene **85**, which was trapped with methyl 2-ethylacrylate (**86**) to yield compound **87**, after final treatment with an excess of allyl bromide (Scheme 35).^[54] Transition state TS was postulated to explain the observed stereochemistry. Compound **87** was easily transformed into different alkaloids, such as (–)-tabersonine, (–)-vincedidifformine or (–)-aspidospermidine, by known methodologies.

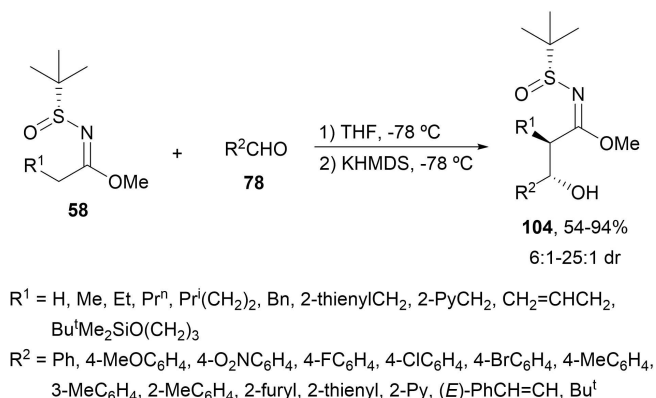
Aldehydes **78** condensed with the lithium salt of amino acid **88** to yield intermediate **I**, which then reacted with imine **12** to yield, in the presence of a catalytic amount of 3-nitrobenzoic acid, diamines **89** (Scheme 36).^[55] The transition state shown in Scheme 36 can explain the stereochemistry observed in compounds **89**.

3.2. Organosodium Compounds

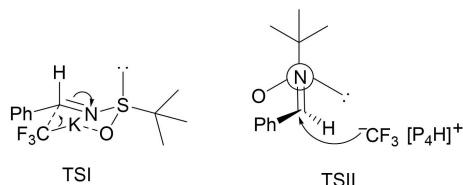
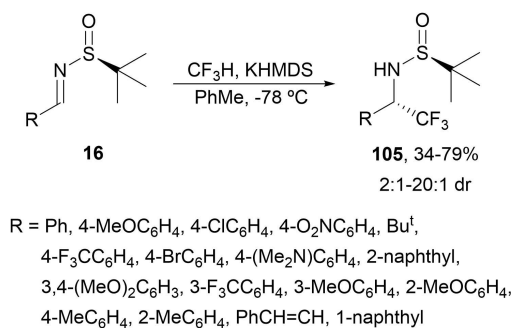
The sodium enolate of compound **90** reacted with imine **12** to yield the corresponding products **91** (Scheme 37).^[56] However by changing the reaction conditions the same reagents gave the



Scheme 43. Reaction of imines **16** with the potassium salt of **102**.



Scheme 44. Reaction of imidates **58** with aldehydes **78** in the presence of KHMDS.



Scheme 45. Reaction of imines **16** with CF_3H and KHMDS.

spiro compounds **92**. For the last transformation, transition state TS and intermediate **I** were proposed to take part in the process.

Imidates **58** react with nitroso arenes in the presence of NaHMDS to give the corresponding products **93** (Scheme 38).^[57] The observed stereochemistry can be explained through the transition state depicted in Scheme 38.

The same starting materials **58** were deprotonated at the α -position by treatment with NaHMDS, followed by oxidation with molecular oxygen in the presence of P(OMe)_3 , to afford compounds **94** (Scheme 39).^[58] The same process was studied with the corresponding 4-morpholinyl derivatives with similar results. The TS shown in Scheme 39 could explain the observed stereochemistry in compounds **94**.

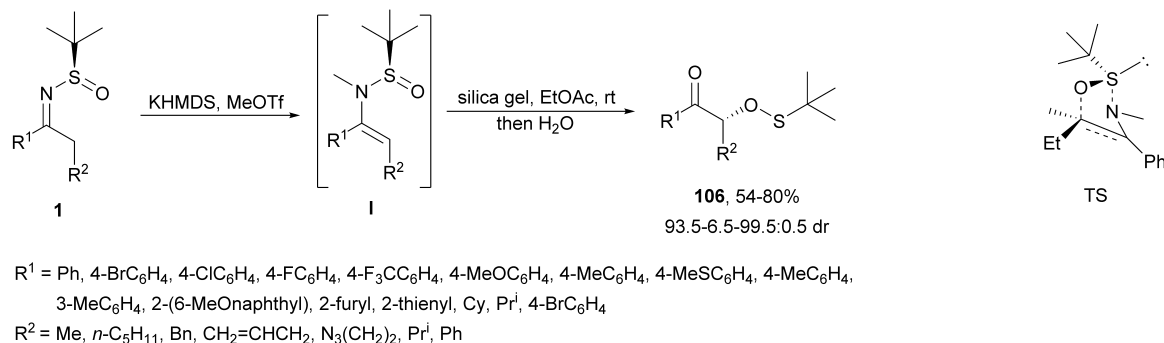
Cyclic sulfinyl enamines **95** were transformed into the corresponding enolates, and alkylated with several R-Br (**96**), to yield the expected products **97** (Scheme 40).^[59] Treatment of compounds **97** with $\text{Pd(PPh}_3)_4$ in morpholine afforded the deprotected products **98**. Transition states TS were proposed to explain the observed stereochemistry.

The reaction of imines **16** with HCF_3 and NaHMDS in THF at -60°C afforded the expected compounds **99** with both good yields and diastereoselectivities (Scheme 41).^[60] When the same reaction was carried out in DMF at -30°C , the corresponding aziridines **100** were isolated. For the preparation of compounds **99**, an approach **I** shown in Scheme 41 would explain the observed stereochemistry.

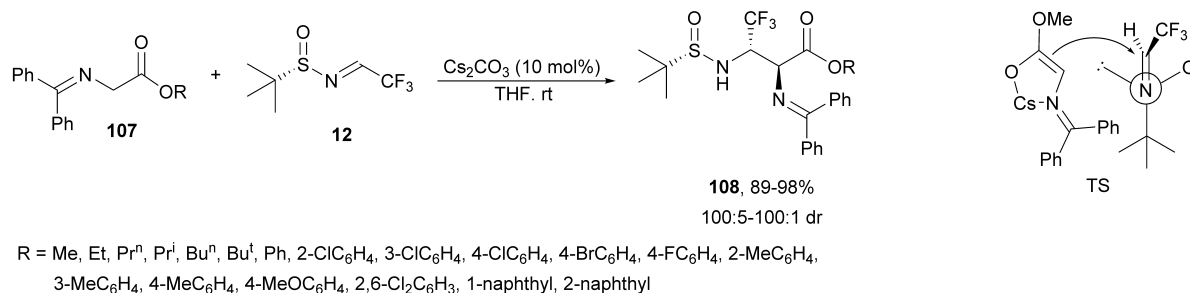
The same type of aziridines **101** were prepared from imines **12** and trimethyl sulfoxonium iodide (TMSOI)-NaH, after nucleophilic addition and cyclizing at -45°C (Scheme 42).^[61] Transition state TS depicted in Scheme 42 has been postulated in order to explain the observed stereochemistry.

3.3. Organopotassium Compounds

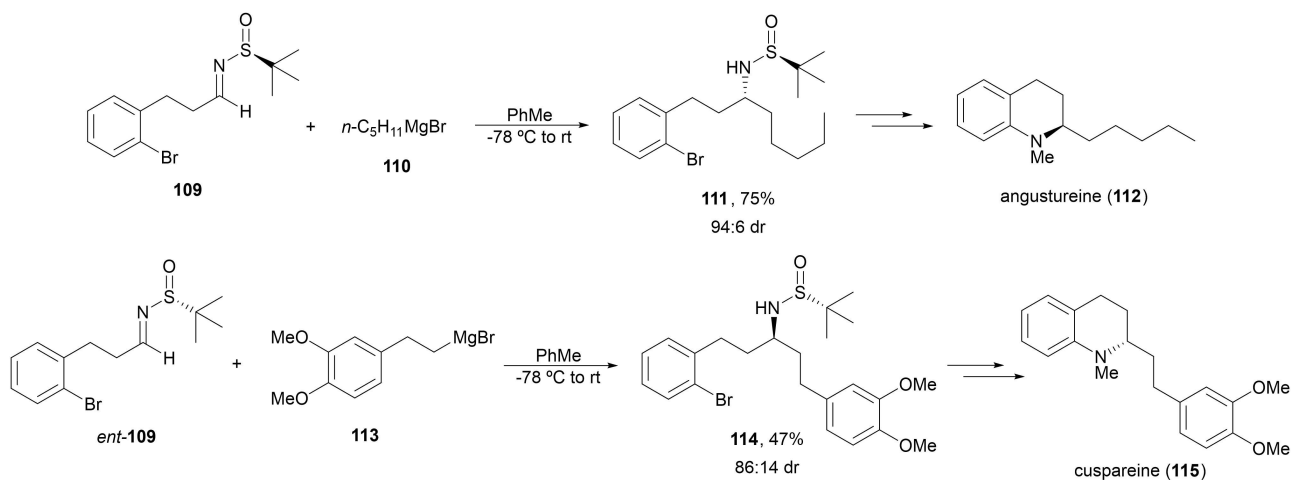
The reaction between imines **16** and the potassium enolate derived from the indole derivative **102** in the presence of $\text{Ti(OPr}^i)_4$ led to the formation of products **103** (Scheme 43).^[62]



Scheme 46. Reaction of ketimines **1** with KHMDS and MeOTf.



Scheme 47. Reaction of imino esters **107** with imine **12**.



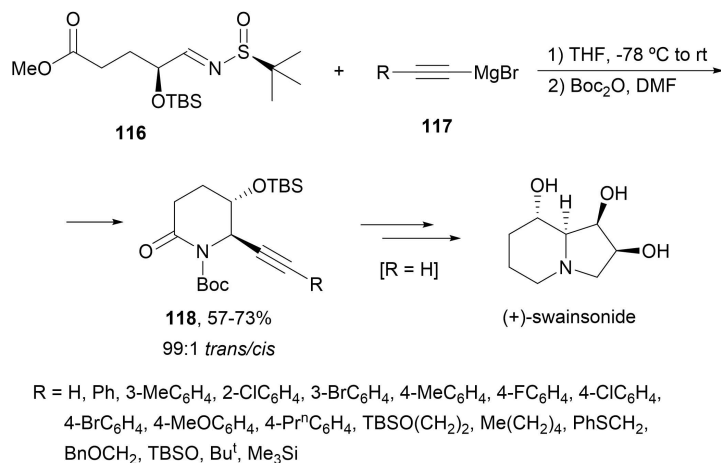
Scheme 48. Reaction of compounds **109** and *ent*-**109** with Grignard reagents **110** and **113**, respectively.

A transition state model that could explain the stereochemistry observed is included in Scheme 43.

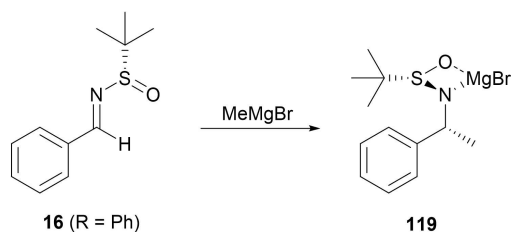
For imidates **58**, their reaction with different aldehydes **78** led to the formation of compounds **104** (Scheme 44).^[63] The

process was also extended to aromatic imidates ($\text{R}^1=\text{Ar}$) with similar results (62–97%, > 20:1 dr).

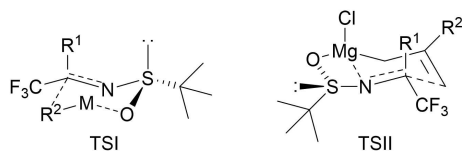
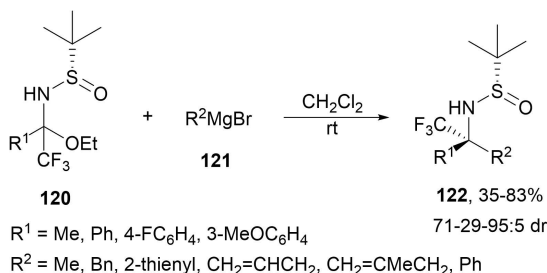
Imines **16** are trifluoromethylated with fluoroform and KHMDS at low temperature, to give the corresponding amines **105** (Scheme 45).^[64] Interestingly, when instead of



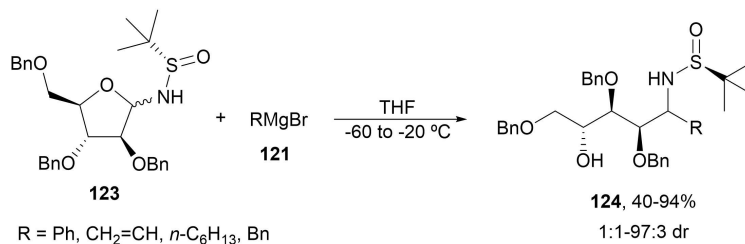
Scheme 49. Reaction of imine **116** with alkynyl Grignard reagents **117**.



Scheme 50. Reaction of imine **16** with MeMgBr.



Scheme 51. Reaction of hemiaminals **120** with Grignard reagents **121**.



Scheme 52. Reaction of glycosylamine **123** with Grignard reagents **121**.

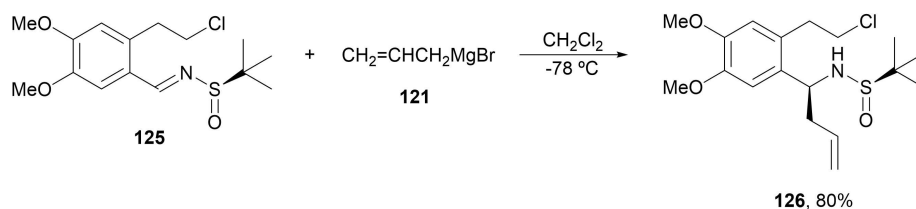
KHMDS a *P*₄-*t*-Bu was used as base, the opposite configuration was obtained for compounds **105** starting from the same materials, and using the same reaction conditions. Transition states TSI and TSII could explain the observed stereochemistry in compounds **105**.

Treatment of ketimines **1** with KHMDS in the presence of MeOTf gives compounds **106** through a [2,3]-rearrangement of **1** and the TS shown in Scheme 46, followed by final hydrolysis with H₂O (Scheme 46).^[65] If the final hydrolysis was performed with hydrochloric acid, the corresponding alcohols were isolated (68–77 %, 99:1 er).

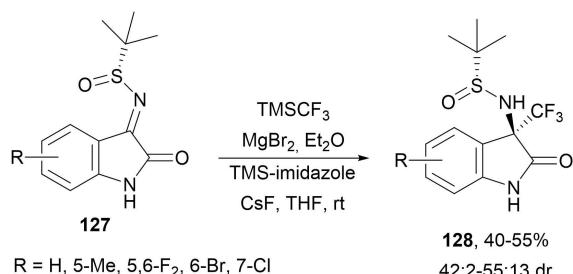
3.4. Organocesium Compounds

The reaction of the cesium enolate of imino esters **107** with imine **12** yielded compounds **108** (Scheme 47).^[66] The non-chelated transition state shown in Scheme 47 can explain the observed stereochemistry in compounds **108**.

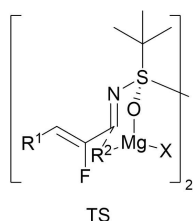
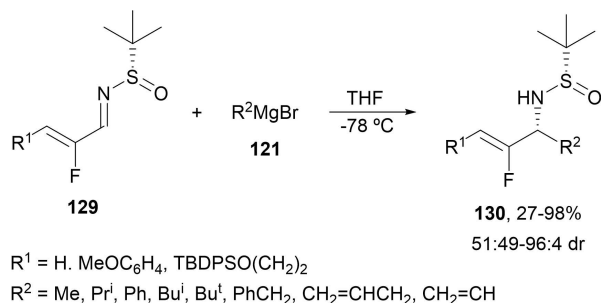
In conclusion, from the results included in this section, we believe that organoalkaline compounds, mainly organolithium derivatives, add diastereoselectively to chiral *N*-*tert*-butylsulfonyl imines to give the expected products. Most or organo-metallic compounds are prepared by deprotonation of the



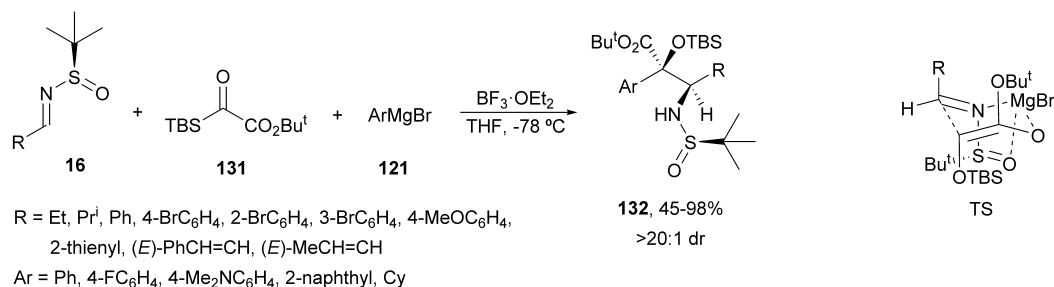
Scheme 53. Reaction of imine **125** with $\text{CH}_2=\text{CHCH}_2\text{MgBr}$ (**121**).



Scheme 54. Reaction of imines **127** with TMSCF_3 and MgBr_2 .



Scheme 55. Reaction of imines **129** with Grignard reagents **121**.



Scheme 56. Reaction of imines **16** with oxoacetate **131** and Grignard reagents **121**.

corresponding starting materials with a metallic base and used after their generation.

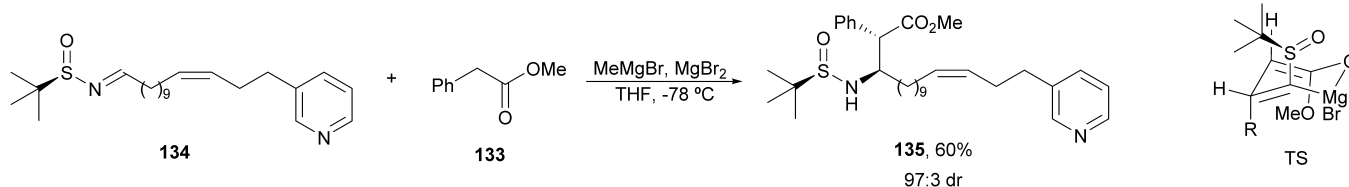
4. Organomagnesium Compounds

The reaction of imine **109** with pentylmagnesium bromide (**110**) led to the formation of compound **111**, which by deprotection-Pd amination-methylation, afforded (–)-angustureine **112** (Scheme 48).^[67] Starting from *ent*-**109** and using 3,4-(MeO)₂C₆H₃MgBr (**113**), following the same reactions, (–)-cuspareine **115** was obtained, through compound **114**.

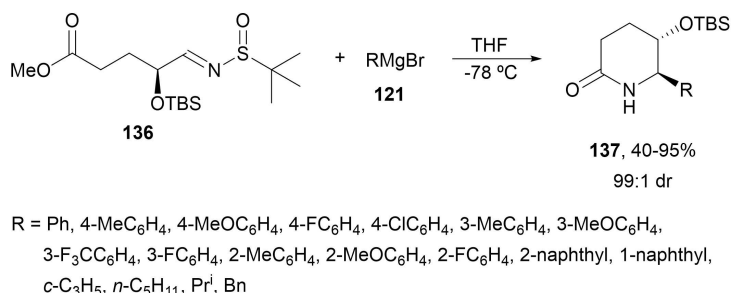
When imine **116** was allowed to react with an alkynylmagnesium bromide **117**, the corresponding product **118** was obtained, after protection with Boc_2O (Scheme 49).^[68] For $\text{R}=\text{H}$, and follow known methodologies, (+)-swainsolide (**119**) was easily obtained. In the reaction of compound **116** with allylmagnesium bromides substituted piperidinones were obtained, which by reported procedures one of the reaction products was transformed into (–)-epiquinamide.

The reaction of imine **16** ($\text{R}=\text{Ph}$) with MeMgBr has been analyzed by DFT calculations, concluding that the (*E*)-**16** gives the corresponding intermediate **119** (Scheme 50).^[69] The results, which are in agreement with the experimental data, were also extended to the same reaction with MeLi .

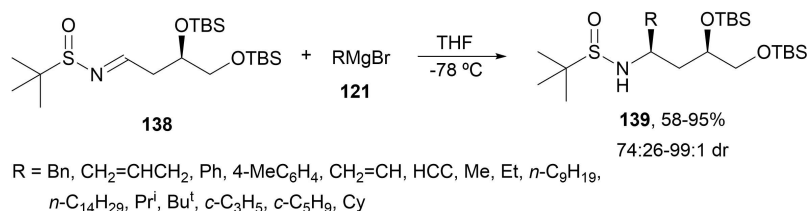
Hemiaminals **120** react with Grignard reagents **121** to give mainly the corresponding products **122** (Scheme 51).^[70] The same reaction was also performed using alkyllithium reagents with similar results. The process seems to take place through the Zimmerman-Traxler six-membered transition states TSI



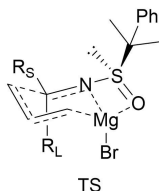
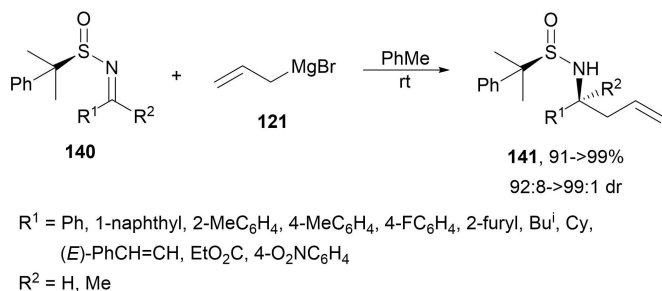
Scheme 57. Reaction of imine **134** with the Mg-enolate of **133**.



Scheme 58. Reaction of imine **136** with Grignard reagents **121**.



Scheme 59. Reaction of imine **138** with Grignard reagents **121**.



Scheme 60. Reaction of imines **140** with allylmagnesium bromide.

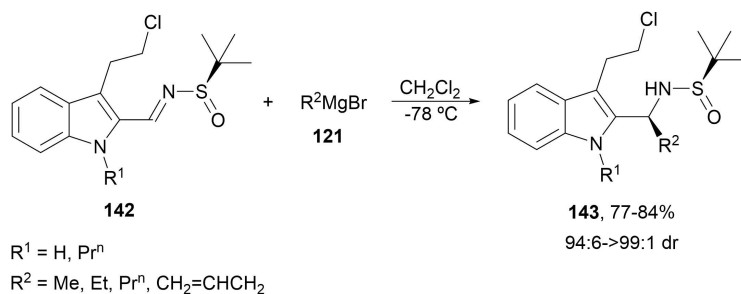
and TSII for the allylation process, with coordination between the sulfinyl oxygen of the intermediate imine and the metal.

In the case of glycosylamine **123** the addition of Grignard reagents **121** led to the formation of compounds **124** (Scheme 52).^[71]

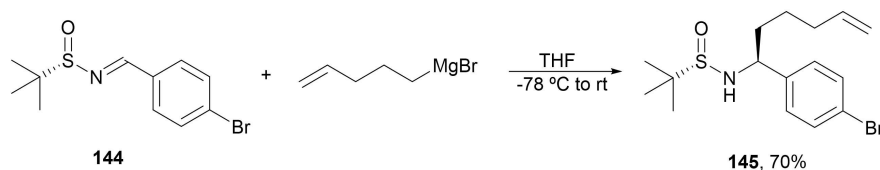
The reaction of imine **125** with allylmagnesium bromide **121** (R=CH₂=CHCH₂) led to the formation of compound **126** (Scheme 53).^[72] This compound was easily transformed into (–)-crispin by known methodologies. When methylmagnesium bromide was used, (–)-salsolidine was prepared after cyclization-dechlorination-desulfonylation.

In the case of imines **127** derived from isatin, their reaction with TMSCF₃ in the presence of MgBr₂ yielded the corresponding products **128** with modest diastereoselectivities (Scheme 54).^[73] The removal of the sulfinyl group under acidic conditions (HCl/MeOH, 0 °C) led to the corresponding amino compounds (R=H) with good results (90 %, 98 % ee).

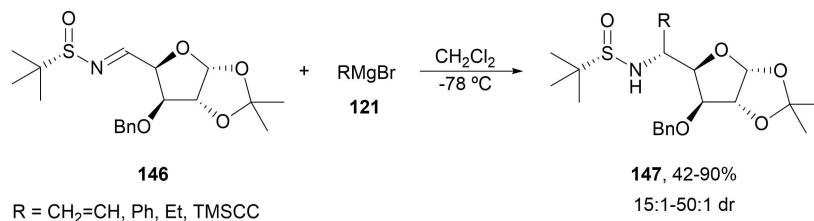
α-Fluoroimines **129** react with Grignard compounds **121** to give the corresponding products **130** (Scheme 55).^[74] The



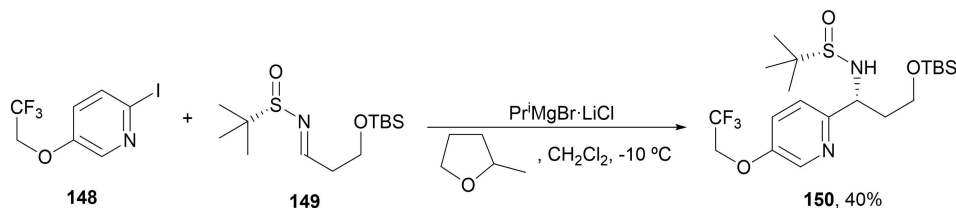
Scheme 61. Reaction of imines **142** with Grignard reagents **121**.



Scheme 62. Reaction of imine **144** with pentenylmagnesium bromide.



Scheme 63. Reaction of imine **146** with Grignard reagents **121**.



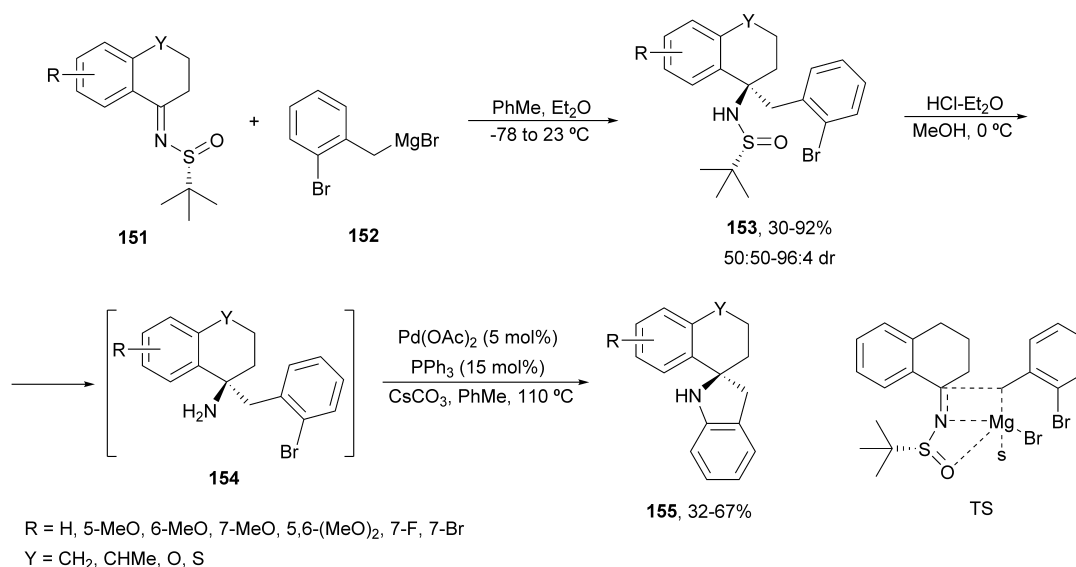
Scheme 64. Reaction of imine **149** with the Grignard reagent derived from **148**.

process can also take place in the presence of ZnMe_2 , so the corresponding zincates will add to the starting material **129** with different results. The open transition state TS shown in Scheme 55 has been proposed in order to explain the obtained results.

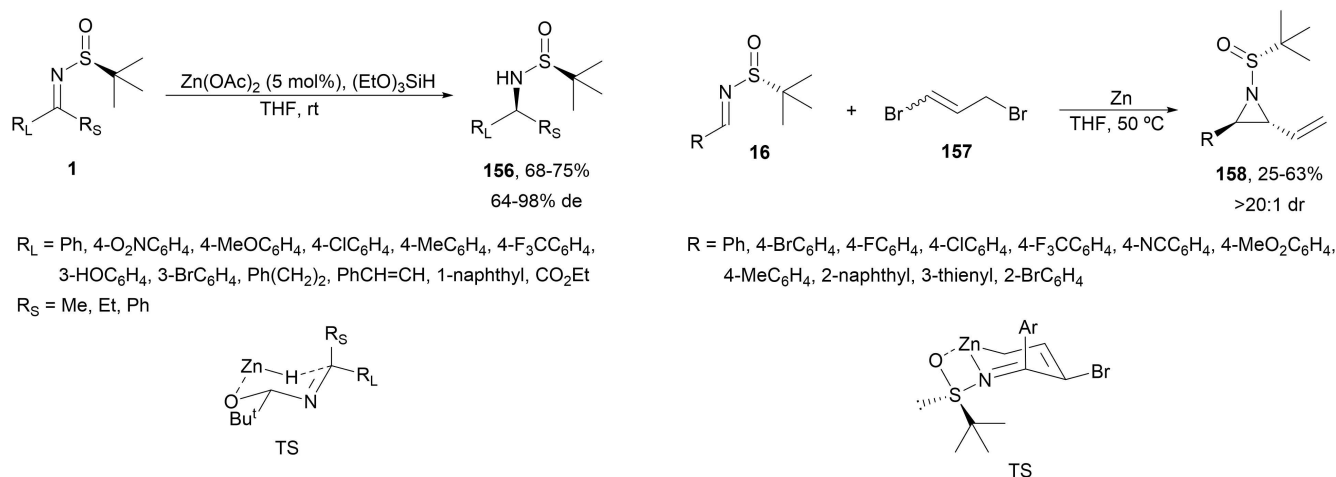
A three-component reaction of imines **16**, the oxoacetate **131**, and a Grignard compound **121**, led to the formation of compounds **132** with high diastereoselectivity (Scheme 56).^[75] However, using an alkyllithium reagent, under similar reaction

conditions, gave a different diastereomer ($2R,3S,R_S$) as the major reaction product (>20:1 dr). A chair-like transition state TS was proposed to take part in the reaction, in order to explain the stereochemistry observed in products **132**.

In the preparation of (–)-(2*S*,3*R*,*Z*)-nakinadine, a key step was the addition of the magnesium enolate of phenylacetic ester **133** to the imine **134**, in the presence of MgBr_2 , to yield compound **135** (Scheme 57).^[76] Transition state TS was postulated to explain the observed stereochemistry.



Scheme 65. Reaction of imines **151** with benzylmagnesium bromide **152**.



Scheme 66. Reaction of imines **1** with (EtO)₃SiH in the presence of Zn (OAc).

Scheme 67. Reaction of imines **16** with 1,3-dibromopropene **157** in the presence of Zn.

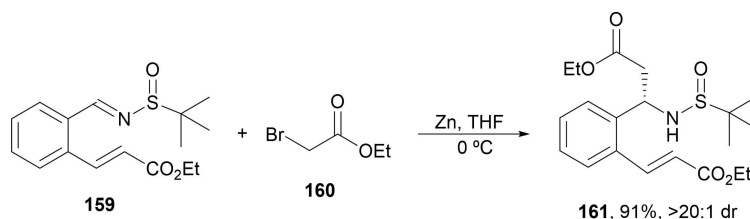
Chiral substituted 2-piperidones **137** were diastereoselectively prepared by addition of Grignard reagents **121** to imines **136** (Scheme 58).^[77] For R=Ph, after standard transformations, compound (+)-CP-99,994 was obtained.

The same group has reported the reaction of imine **138** with Grignard reagents **121** to give mainly the corresponding products **139** (Scheme 59).^[78] After monodeprotection, compounds **139** were transformed into chiral pyrrolizidines by oxidation with PDC or CrO₃. The reaction can be also carried out starting from compounds **136** (Scheme 58) under Barbier conditions (Mg, RX) giving directly the corresponding substituted pyrrolidones.^[79]

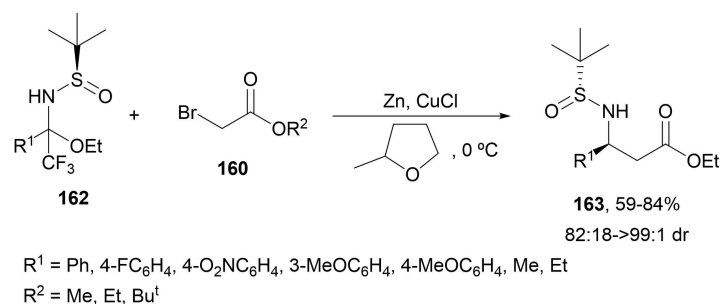
The reaction of imines **140** with allylmagnesium bromide **121** (R=CH₂=CHCH₂) led to the formation of compounds **141** with excellent diastereoselectivities (Scheme 60).^[80] The observed stereochemistry can be easily explained considering the transition state TS shown in Scheme 60.

For the preparation of a series of tetrahydro-β-carboline alkaloids, imine **142** reacts with different Grignard reagents **121** to yield the expected compounds **143** with excellent diastereoselectivity (Scheme 61).^[81] After cyclization (NaH, DMF), the corresponding tricyclic compounds were obtained.

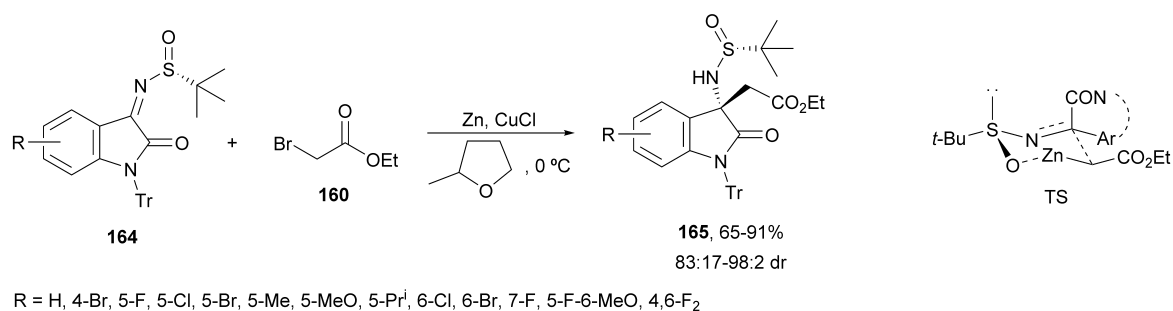
Imine **144** was treated with pentenylmagnesium bromide to afford compound **145** (Scheme 62).^[82] This product was



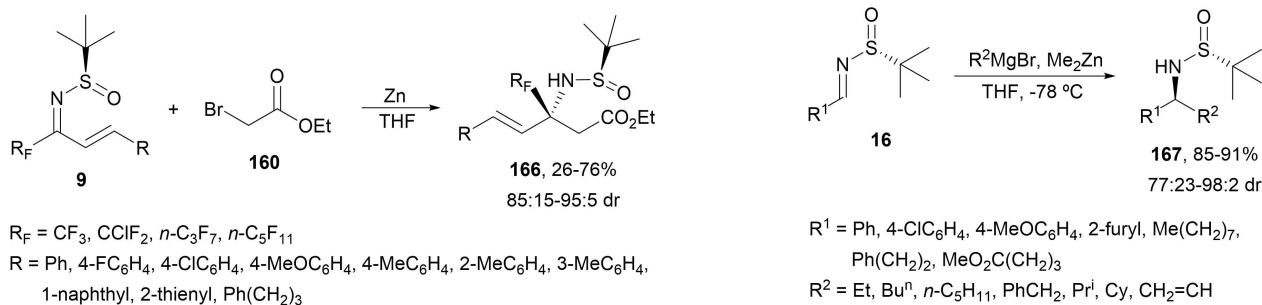
Scheme 68. Reaction of imine **159** with the Reformatsky reagent.



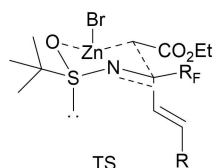
Scheme 69. Reaction of imine precursors **162** with the Reformatsky ester **160**.



Scheme 70. Reaction of imines **164** with the Reformatsky reagent.



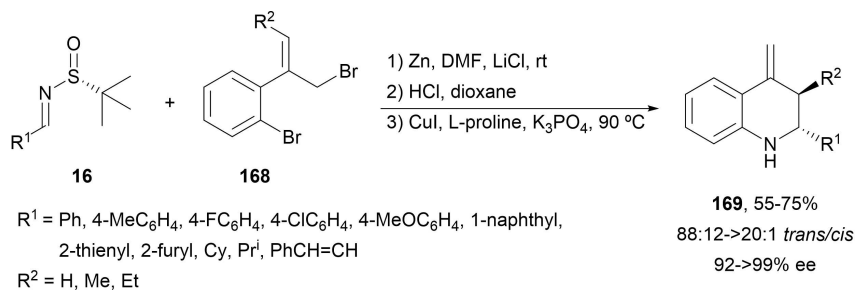
Scheme 72. Reaction of imines **16** with *in situ* generated zincates.



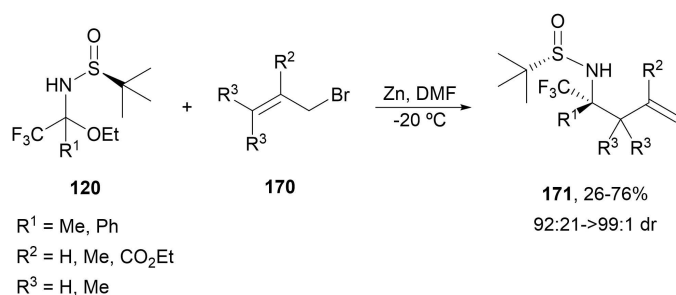
Scheme 71. Reaction of imine **9** with the Reformatsky reagent.

then transformed into the corresponding *trans*-disubstituted piperidines by conventional methodologies.

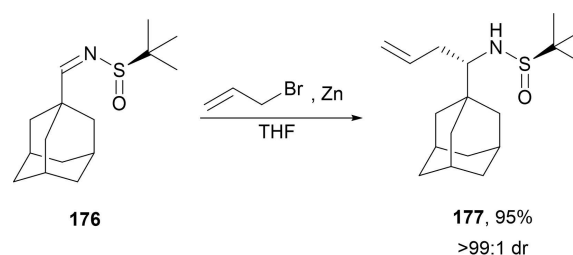
When D-xylose derived imine **146** was allowed to react with different Grignard reagents **121**, the corresponding



Scheme 73. Reaction of imines **16** with allyl bromide **168** and Zn.



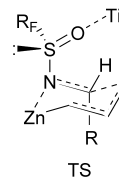
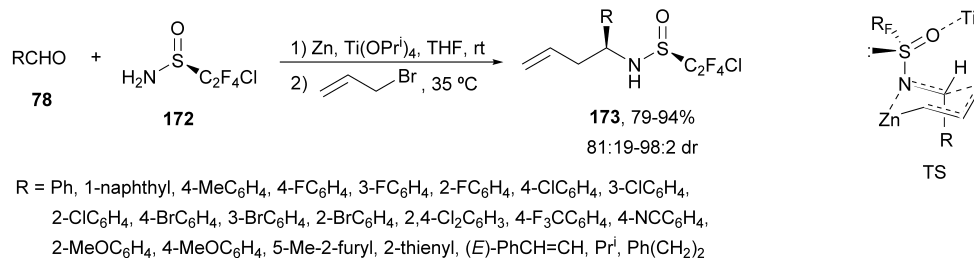
Scheme 74. Reaction of compounds **120** with allyl bromides **170** and Zn.



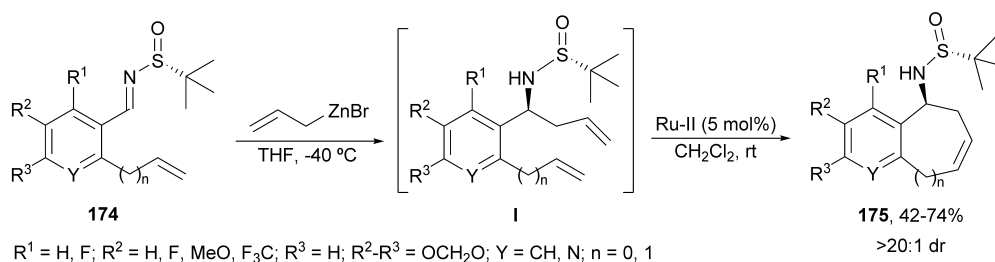
Scheme 77. Reaction of adamantyl imine **176** with *in situ* generated allylzinc bromide.

products **147** were isolated with excellent diastereoselectivities (Scheme 63).^[83]

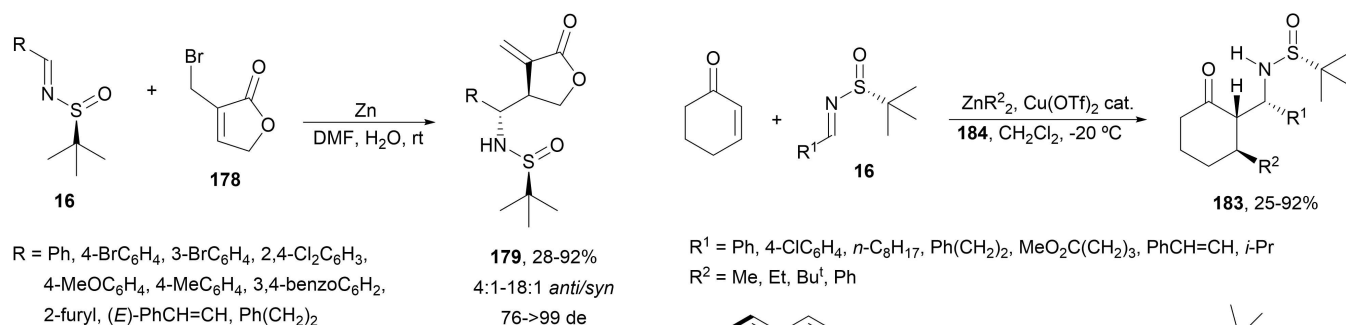
In situ generated arylmagnesium halide from aryl iodide **148** could be transformed into the corresponding product **150** by reaction with imine **149** (Scheme 64).^[84]



Scheme 75. Reaction of *in situ* generated imines with allyl bromide and Zn.



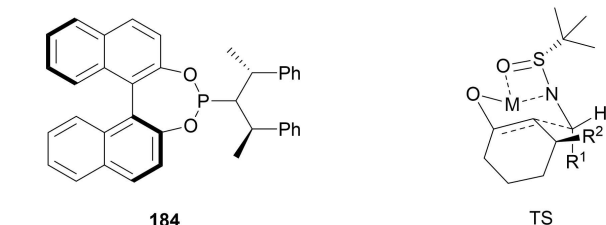
Scheme 76. Reaction of imines **174** with allylzinc bromide following by RCM.



Scheme 78. Reaction of imines **16** with the organozinc compound derived from **178**.

The reaction of imines **151** with *o*-bromobenzylmagnesium bromide **152** led to the formation of the expected compounds **153** (Scheme 65).^[85] Desulfonylation of **153** yielded compound **154**, which were cyclized under a Pd-catalyzed process, giving compounds **155**. The same reactions were performed with imines *ent*-**151**, so after the whole process, compounds *ent*-**155** were obtained. Transition state TS depicted in Scheme 65 could explain the obtained results.

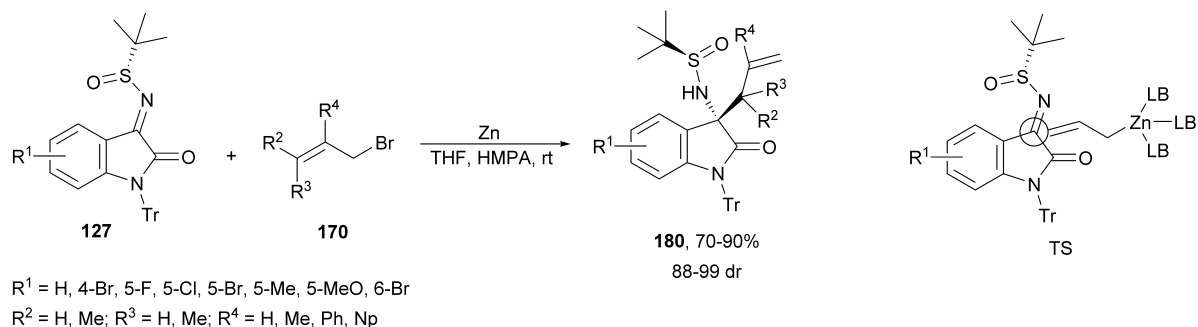
Finally, the magnesium enolate derived from an isomerization of allylic alcohols, and their reaction with imines **16**,



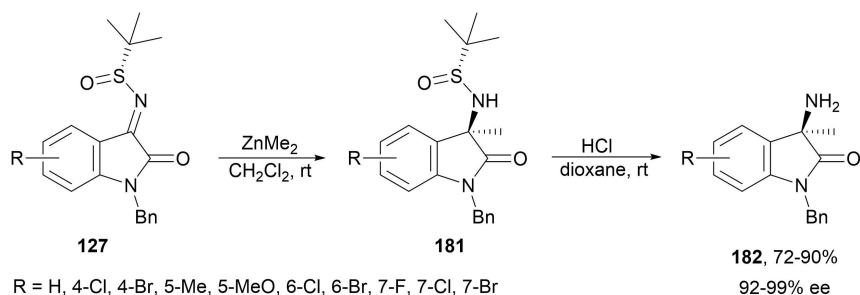
Scheme 81. Reaction of imines **16** with cyclohexanone in the presence of zinc reagent.

yielded the expected products in the presence of NiHCl(dppe)/MgBr₂ in THF at −50 to −40 °C.^[86]

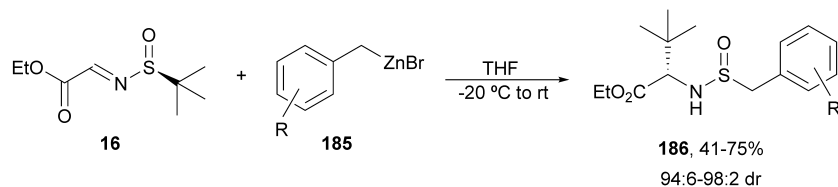
From the results commented in this section, we conclude that Grignard compounds add diastereoselectively to imines,



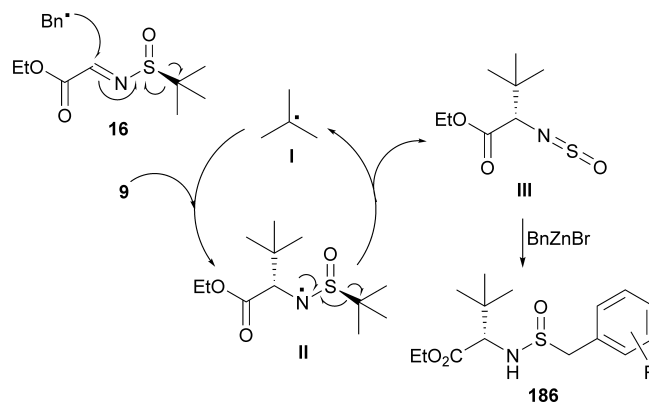
Scheme 79. Reaction of imine **127** with the organozinc reagent derived from **170**.



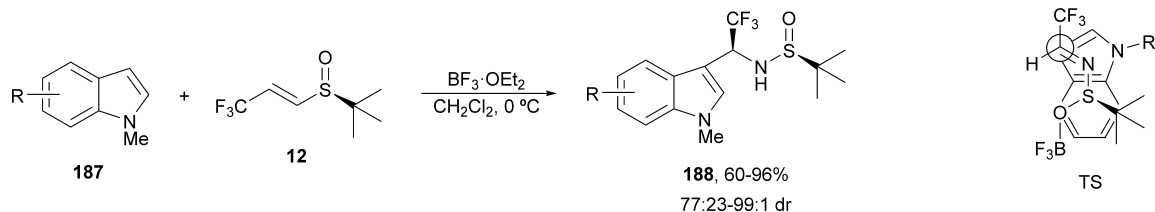
Scheme 80. Reaction of imines **127** with ZnMe₂.



R = H, 2-F, 3-F, 4-F, 2-Cl, 3-Cl, 4-Cl, 4-Br, 3-CN, 2-Me, 4-Me, 3-CF₃, 4-CF₃, 4-CO₂Et

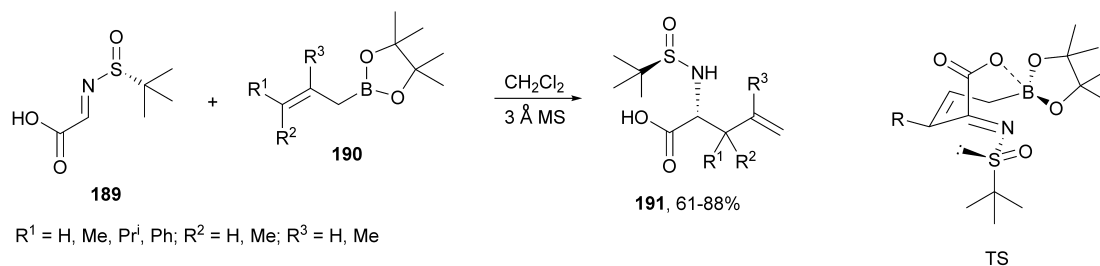


Scheme 82. Reaction of imine **16** with benzylzinc reagents.



R = 2-Me, 2-CHO, 4-CN, 4-MeO, 5-CN, 5-MeO, 5-CO₂Me, 5-CO₂H, 6-Cl, 6-Br, 7-Br, 7-Me, 2-Me, 5-MeO

Scheme 83. Reaction of indoles **187** with imine **12**.

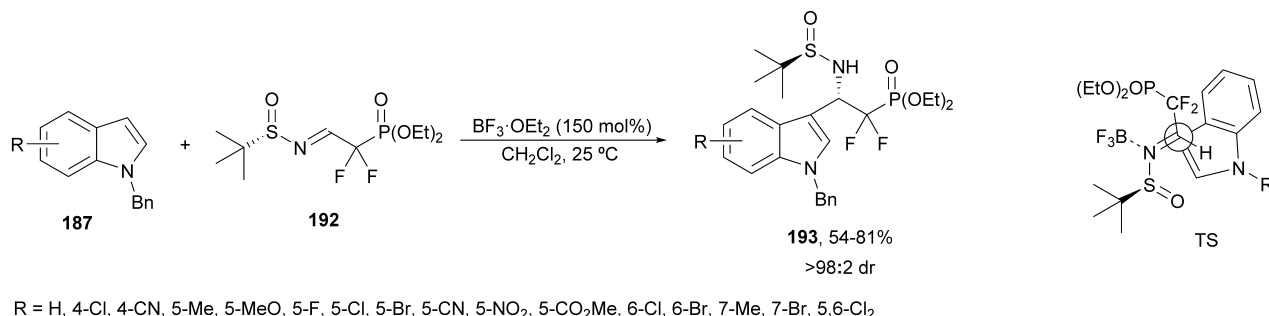


Scheme 84. Reaction of imine **189** with allyl boranes.

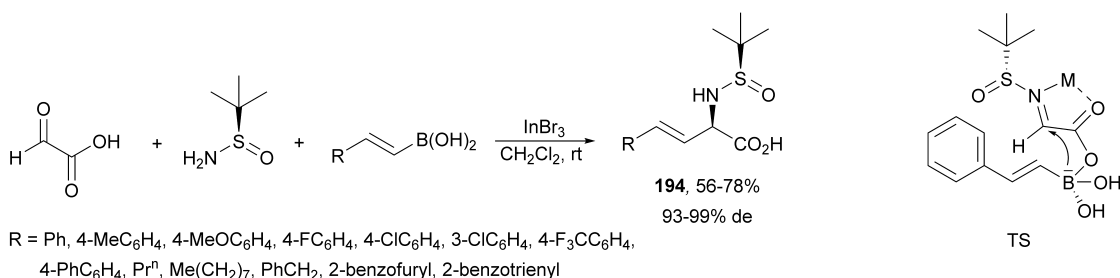
to give the corresponding alkylated or arylated products. In some cases, magnesium enolates are used in the reaction with imines.

5. Organozinc Compounds

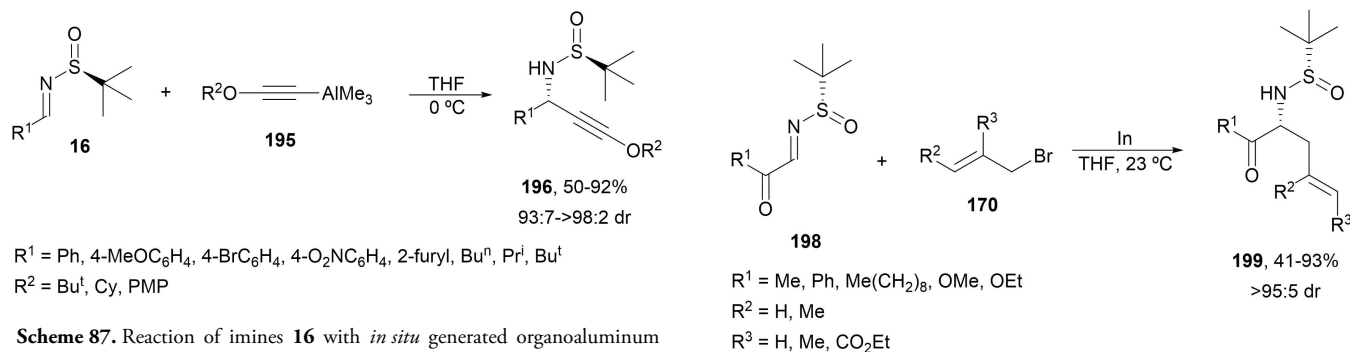
Zn acetate catalyzed the reduction of ketimines **1** with (EtO)₃SiH to yield the corresponding amines **156**



Scheme 85. Reaction of indoles **187** with imine **192**.

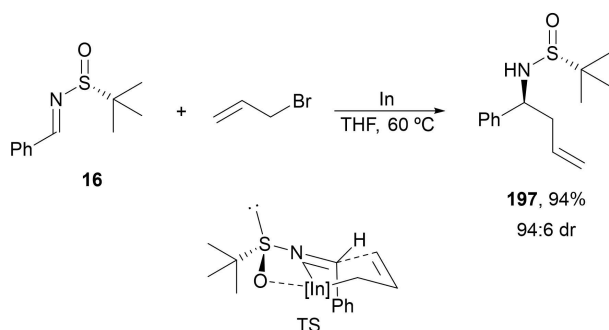


Scheme 86. Three-component reaction of *in situ* generated imines **189** (Scheme 84) with vinyl boronic acids.



Scheme 87. Reaction of imines **16** with *in situ* generated organoaluminum compounds **195**.

Scheme 89. Allylation of compounds **198**.



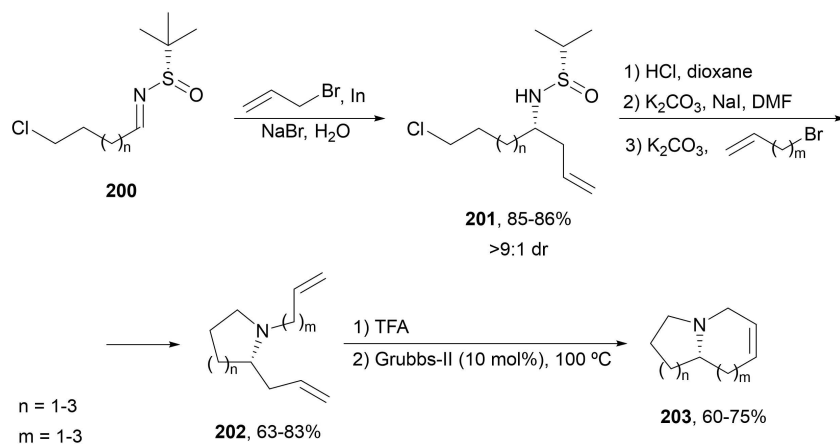
Scheme 88. Allylation of imine **9** in the presence of In.

(Scheme 66).^[87] A transition state TS would explain the obtained stereochemistry.

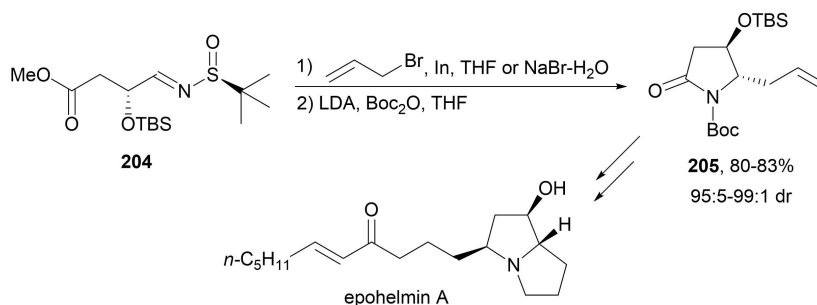
In a Barbier-type reaction, 1,3-dibromopropene **157** add to imines **16** in the presence of Zn to yield the corresponding aziridines **158** (Scheme 67).^[88] Intermediate **I** has been proposed to explain the stereochemistry outcome.

The Reformatsky reaction of imine **159** with the ester **160** led to the formation of sulfinamide derivative **161** (Scheme 68).^[89]

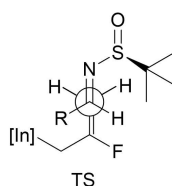
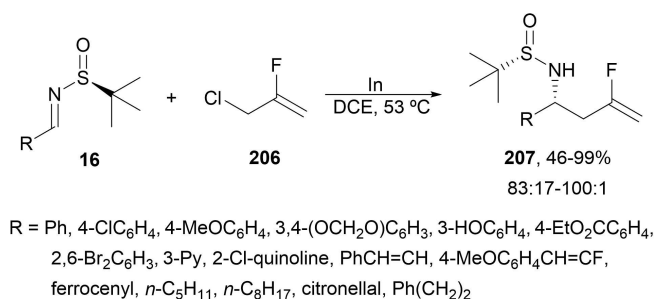
The same reaction was explored with imine precursors **162** to give the corresponding amines **163** (Scheme 69).^[90] Once



Scheme 90. Alkylation of imines **198** and further transformations.



Scheme 91. Alkylation of imine **204**.



Scheme 92. Alkylation of imines **16** with fluoroallyl chlorides **206**.

compounds **163** were hydrolyzed to give the corresponding protected amino acids, their coupling to alanine or phenyl-

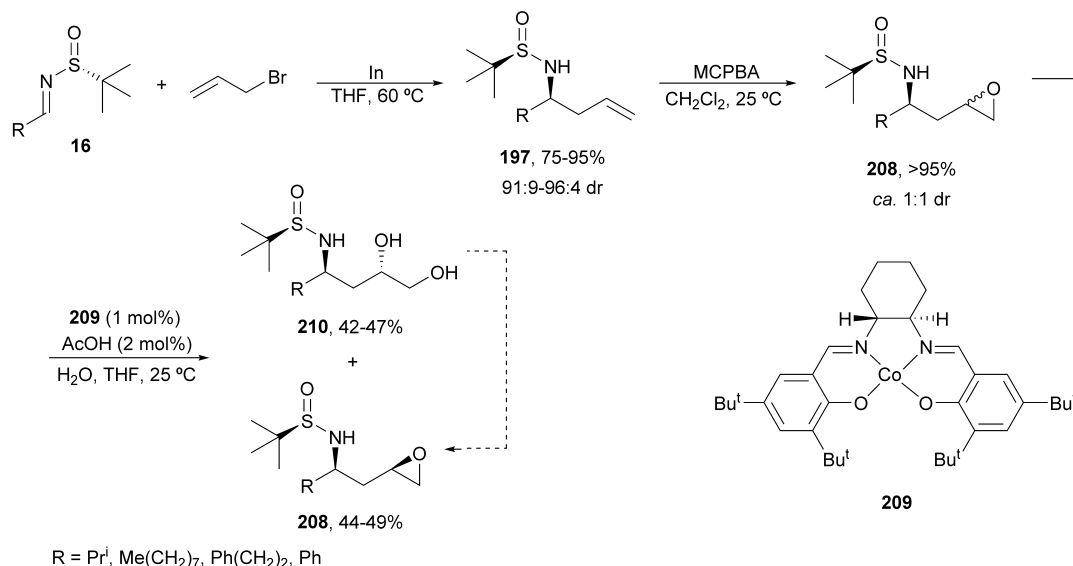
alanine yielded the expected dipeptides with excellent results (93–96% yields).

The Reformatsky reaction applied to imines **164** led to the formation of the corresponding amines **165** (Scheme 70).^[91] The proposed transition state TS can explain the observed stereochemistry in products **165**.

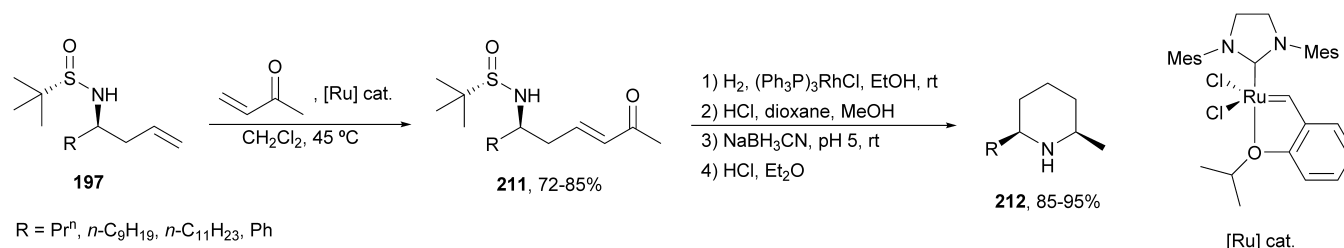
The same reaction applied to imines **9** led to the formation of amines **166** with high diastereoselectivity (Scheme 71).^[92] Transition state TS was proposed to explain the stereochemical outcome of the process.

The addition of *in situ* generated zincates to imines **16** led to the formation of the corresponding amines **167** with high diastereoselectivity (Scheme 72).^[93,94] The reaction was also extended to ketimines **1** with similar results. It is remarkable that the Me group is not transfer to the imine. On the other hand, sulfonamides **167** were hydrolyzed to the corresponding amines with hydrochloric acid in MeOH with good results (45–82%, 54–96% ee).

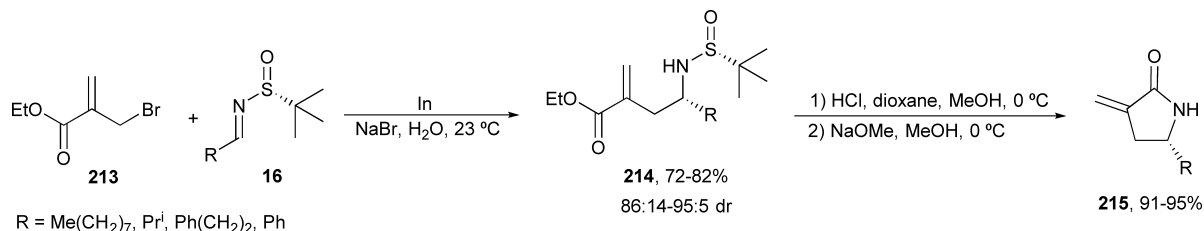
The reaction shown in Scheme 72 was applied to the preparation of chiral amino acids starting from furfural derived imines **16**, or vinyl or allyl zincates.^[95]



Scheme 93. Alkylation of imines **16** followed by oxidation.



Scheme 94. Preparation of 2,6-disubstituted piperidines **212**.



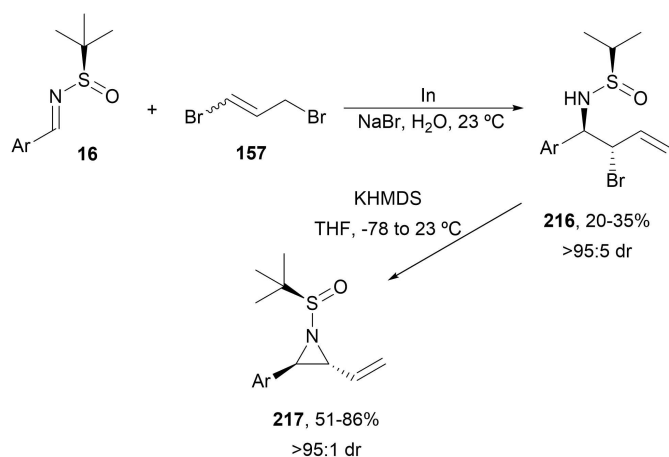
Scheme 95. Reaction of imines **16** with allylic bromide **213**.

Allylations are other type of reactions promoted by Zn. In this sense, the reaction of imines **16** with allyl bromides **168** promoted by Zn yielded products **169**, after desulfonylation and a C–N forming process (Scheme 73).^[96]

Compounds **120**, precursors of the corresponding imines, were allylated with allyl bromides **170** in the presence of Zn, to give the expected products **171** (Scheme 74).^[97] Either, a chelation-controlled TSI in THF, or a steric-controlled TSII

in DMF have been proposed to explain the observed stereochemistry. In some cases the reaction was also studied using In metal.

In situ generated imines derived from aldehydes **78** and sulfonamides **172** were allylated with allyl bromide to produce compounds **173** with high diastereoselectivity (Scheme 75).^[98] TS shown in Scheme 75 could explain the observed stereochemistry.



Ar, 4-FC₆H₄, 4-ClC₆H₄, 4-F₃CC₆H₄

Scheme 96. Reaction of imines **16** with **157**, and further cyclization.

The allylation of imine **174** with allylzinc bromide was followed by Ru-II catalyzed ring-closing metathesis (RCM) to yield bicyclic compounds **175** *via* intermediate **I** (Scheme 76).^[99] Desulfinylation of some compounds **175** gave

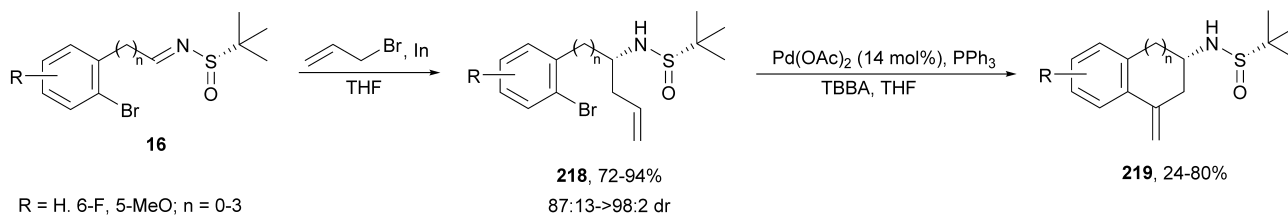
the expected amines, isolated as their hydrochlorides (94–96 % yield).

The same reaction applied to the imine derived from adamantyl aldehyde **176** gave the expected homoallyl amine **177**, which was further desulfinylated to give the expected amine, isolated as its Boc derivative (Scheme 77).^[100]

Another allyl-type starting material for the addition to imines **16** are derived from furanone **178**, so the reaction with Zn in aqueous DMF led to the formation of products **179** (Scheme 78).^[101]

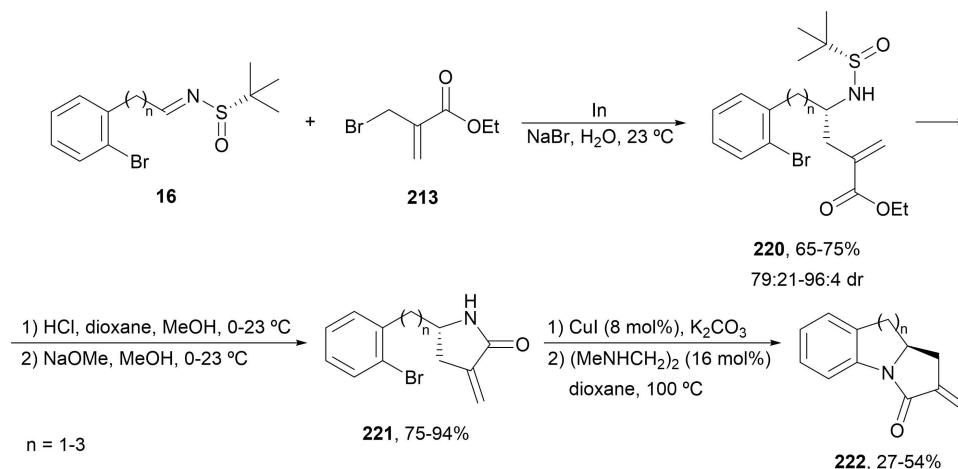
Compound **127** was allylated using an allyl bromide **170** and Zn, affording the corresponding products **180** with high stereoselectivities (Scheme 79).^[102] By using reported methodologies [*N*-allylation (NaH, allyl bromide) and RCM (Ru-II)], compounds **180** were transformed into interesting spirocyclic oxindoles. TS depicted in Scheme 79 is able to explain the stereochemistry observed in compounds **180**.

Similar imines **127** react with Me₂Zn to give the corresponding compounds **181**, which were *in situ* desulfinylated to yield the expected amines **182** (Scheme 80).^[103] The same reaction in the presence of a terminal alkyne introduced the alkynyl group [as (R–C≡C)₂Zn] in the molecule of the starting imine.

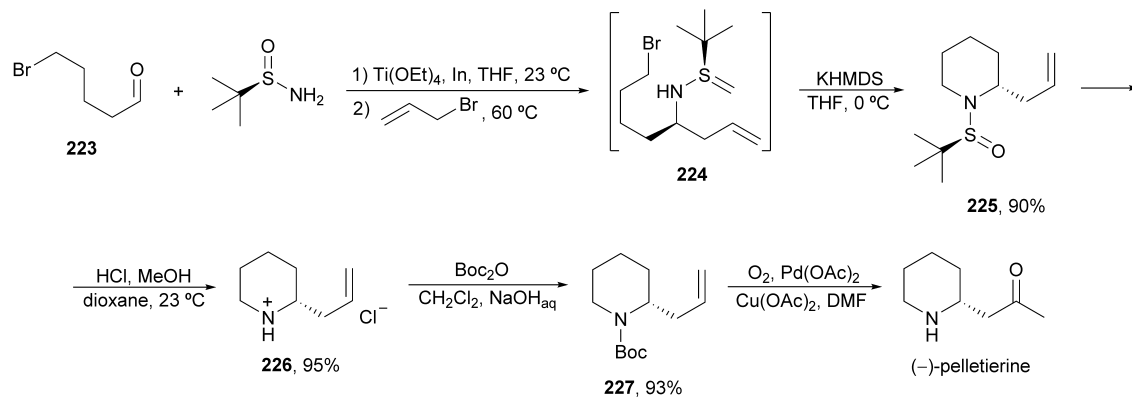


R = H, 6-F, 5-MeO; n = 0–3

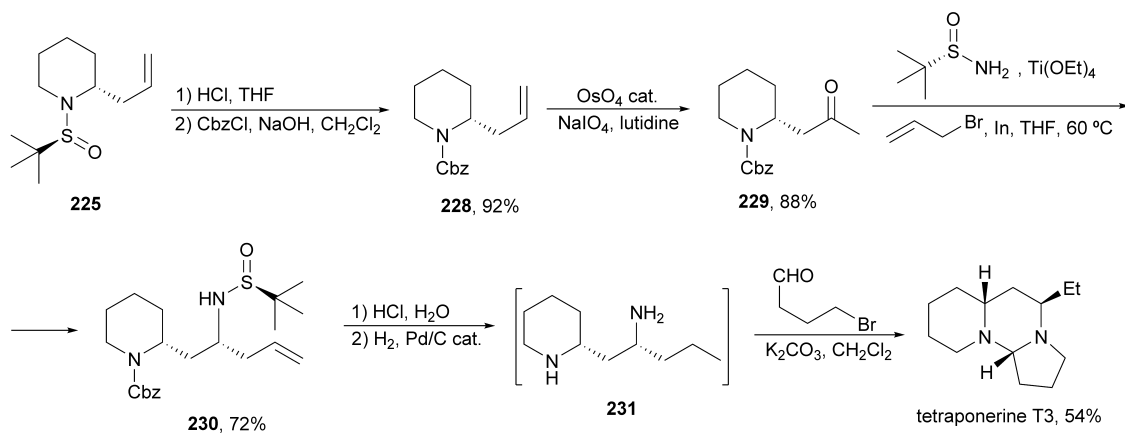
Scheme 97. Allylation of imines **16** and further Pd-catalyzed cyclization.



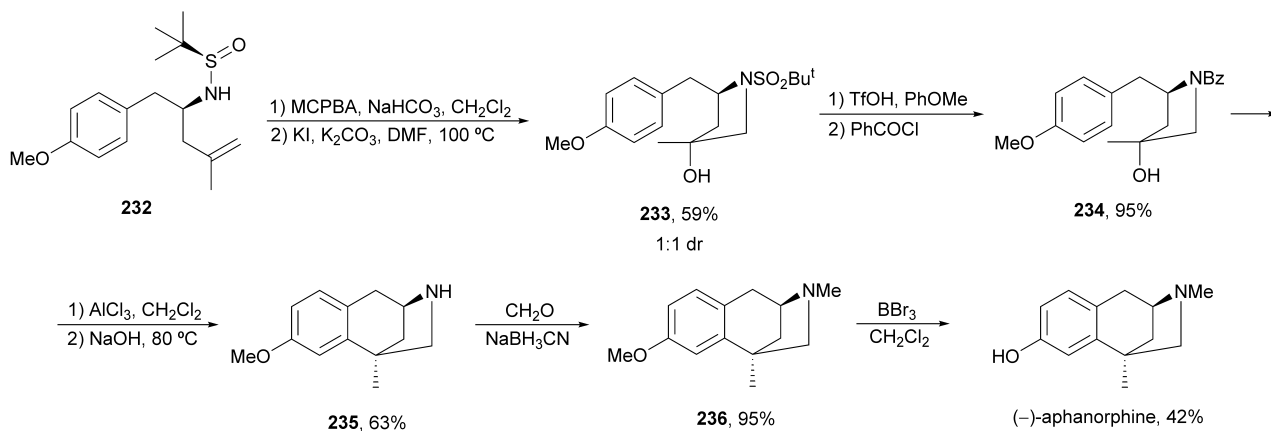
Scheme 98. Reaction of imines **16** with esters **222** and further transformations.



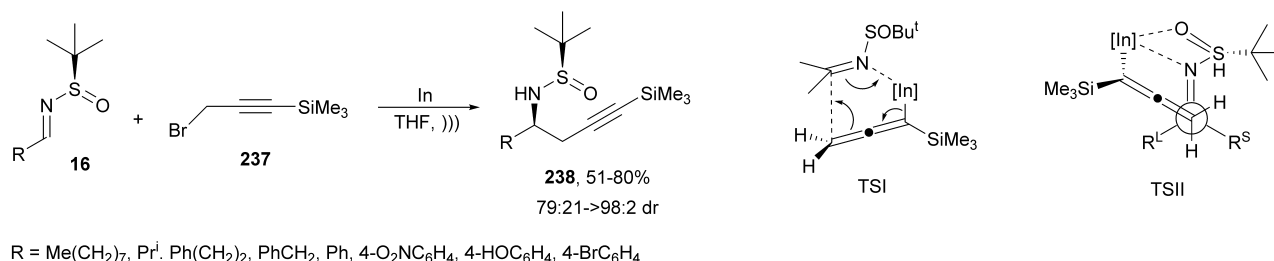
Scheme 99. Preparation of allylpiperidine and further transformation.



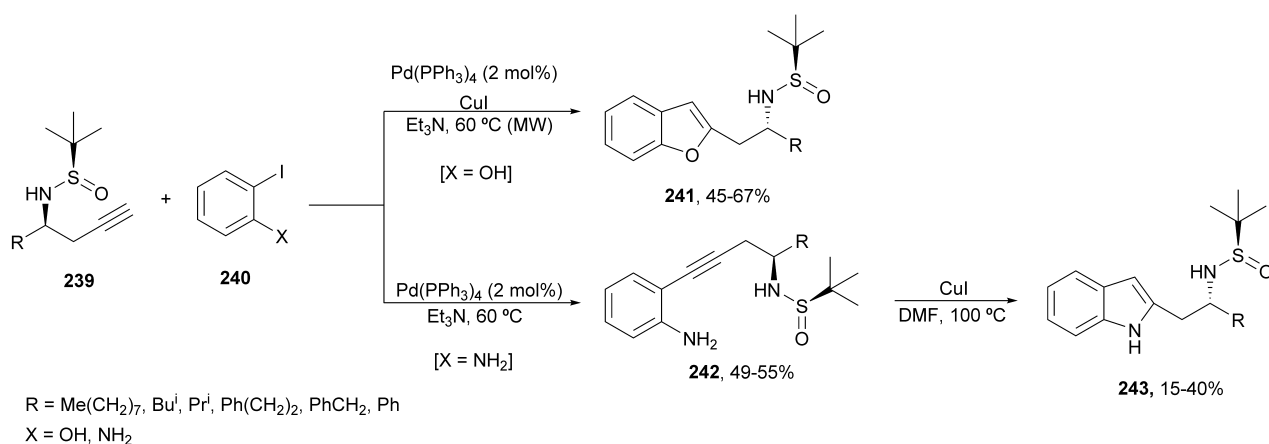
Scheme 100. Preparation of tetraopnerine T3.



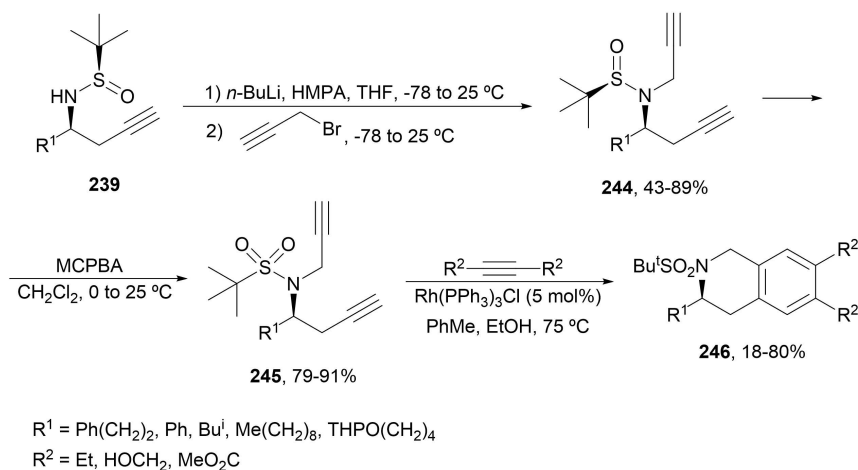
Scheme 101. Synthesis of (–)-aphanorphine.



Scheme 102. Propargylation of imines **16**.



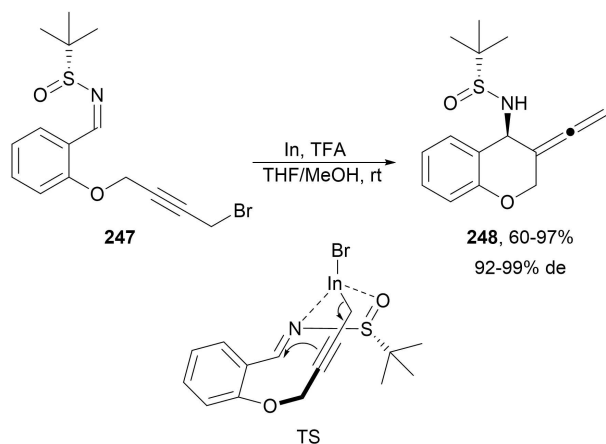
Scheme 103. Reaction of amines **239** with compounds **240**.



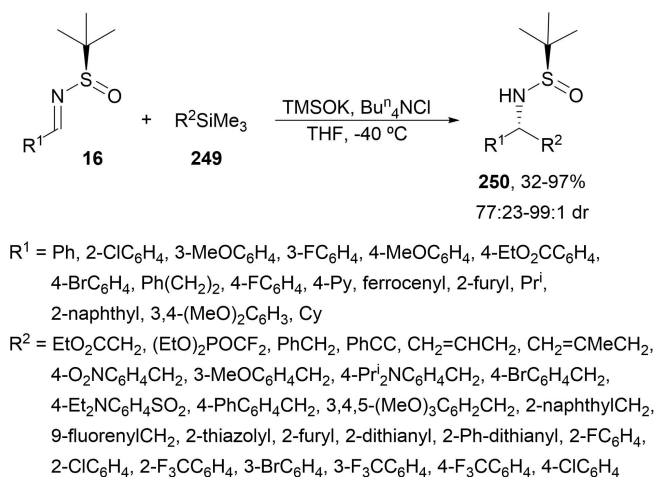
Scheme 104. Propargylation of **239** and further transformations.

Zinc enolates derived from cyclohexanone react with imines **16** to yield the corresponding products **183** in the presence of $\text{Cu}(\text{OTf})_2$, and the ligand **184** (Scheme 81).^[104] The reaction was also extended to cyclopentenone and cyclo-

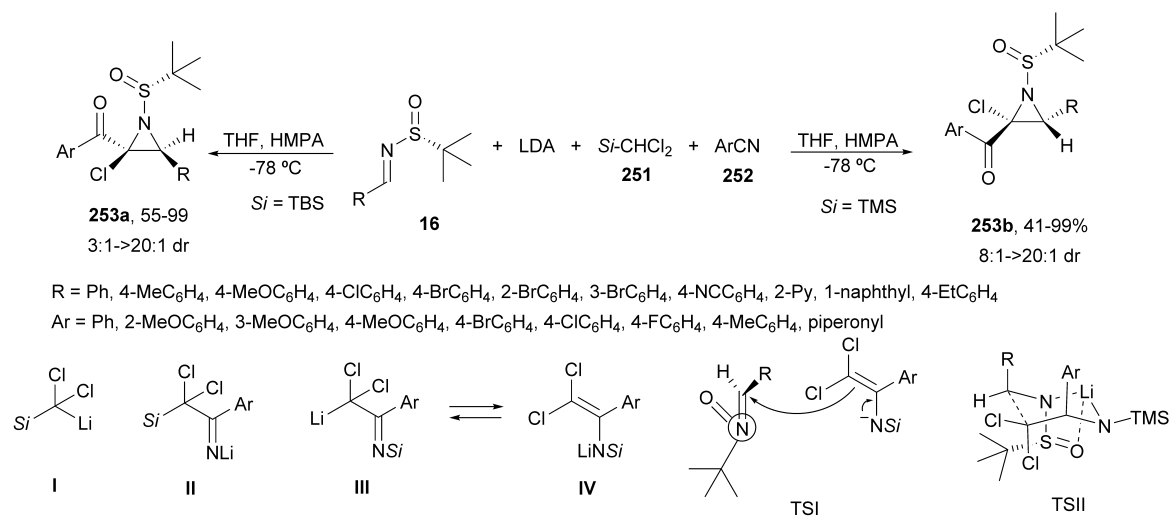
heptenone with similar results. Varying the structure of the sulfonamide and the ligand, all four diastereomers were obtained with excellent diastereoselectivity. Transition state TS would explain the obtained results in terms of stereochemistry.



Scheme 105. Intramolecular cyclization of compound **247**.



Scheme 106. Reaction of imines **16** with organosilicon compounds **249**.



Scheme 107. Cascade reaction of imines **16** with the lithium derivative from **251** and a nitrile **252**.

Finally, imines **16** are allowed to react with benzylic organozinc compounds **185** to afford the corresponding products **186** as consequence of a migration of the Bu^t group from the S atom to the C=N bond (Scheme 82).^[105] As a plausible mechanism, the one depicted in Scheme 82 involving intermediates **I–III** has been proposed to explain the migration of the Bu^t group.

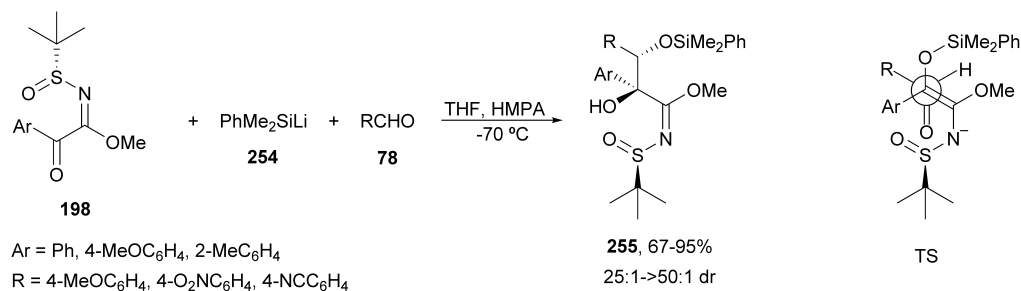
From the results included in this Section, we conclude that organozinc compounds, especially Reformatsky intermediates, allylic and benzylic organozinc intermediates, either isolated or generated *in situ*, are excellent organometallic intermediates to be added to *N*-*tert*-butylsulfinyl imines in a diastereoselective manner.

6. Organoboron Compounds

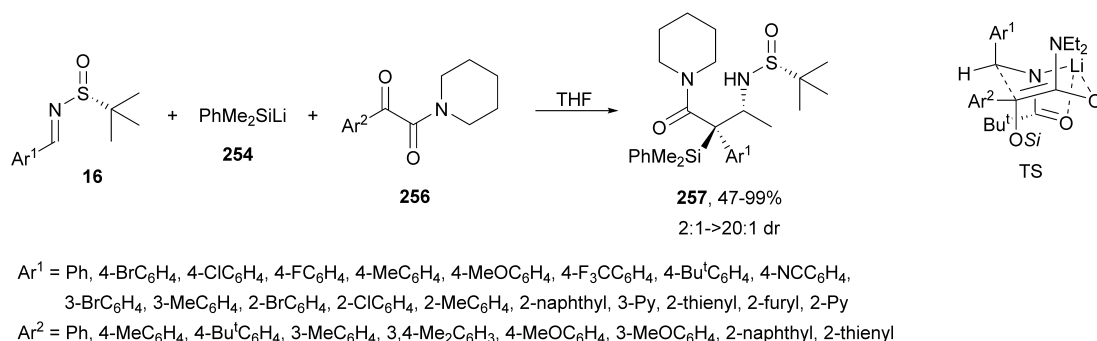
The reaction of indoles **187** with imines **12** in the presence of BF₃·OEt₂ afforded the corresponding products **188** (Scheme 83).^[106] Transition state TS depicted in Scheme 83 can explain the observed stereochemistry in compounds **188**.

In situ generated imines **189** reacted with organoboron allyl derivatives **190** to provide the corresponding products **191** (Scheme 84).^[107] Compounds **191** were easily desulfinylated with HCl/dioxane to yield the expected α-amino acids with good yields (69–93 %) and diastereoselectivities (94–> 98 % ee). Transition state TS shown in Scheme 84 can explain the observed stereochemistry in compounds **191**.

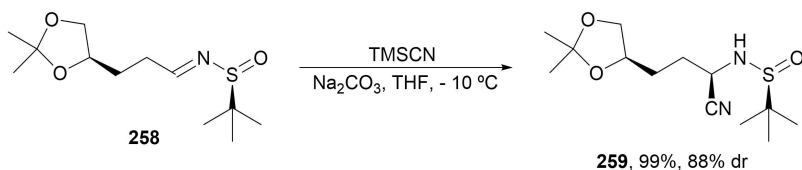
The chemistry depicted in Scheme 83 was also extended to imine **192** having a phosphonate unit, so the corresponding compounds **193** were obtained (Scheme 85).^[108] Also in this case, transition state TS would explain the observed stereochemistry in products **193**.



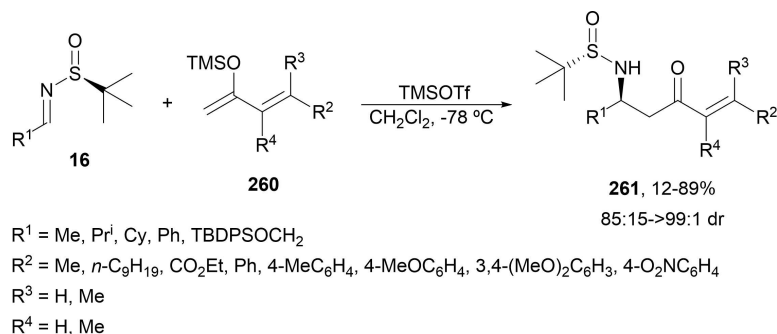
Scheme 108. Reaction of imines **198** with compound **254** and an aldehyde **78**.



Scheme 109. Reaction of imines **16** with organolithium **254** and amides **256**.



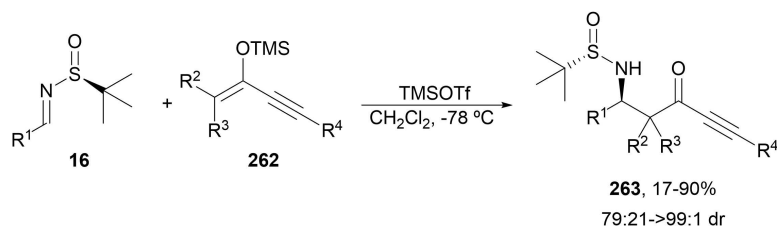
Scheme 110. Reaction of imine **258** with TMSCN.



Scheme 111. Reaction of imines **16** with silyl enol ethers **260**.

Finally, the three-component reaction between glyoxalic acid, a vinyl boronic acid and a sulfinamide in the presence of

InBr₃ led to the formation of the expected protected α -amino acids **194** through the *in situ* generated compounds **189**



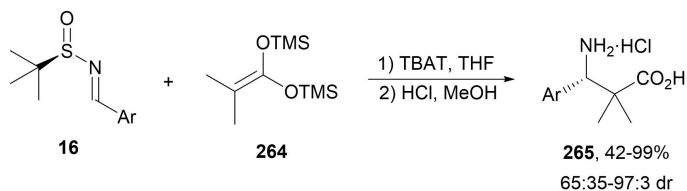
R¹ = TBDPSOCH₂, *n*-C₉H₁₉, Me, Bn, Prⁱ, Br(CH₂)₄, BnOCH₂, EtO₂C, Ph

R² = H, Me

R³ = H, Me, CH₂=CH(CH₂)₂, BnO(CH₂)₃, BnO

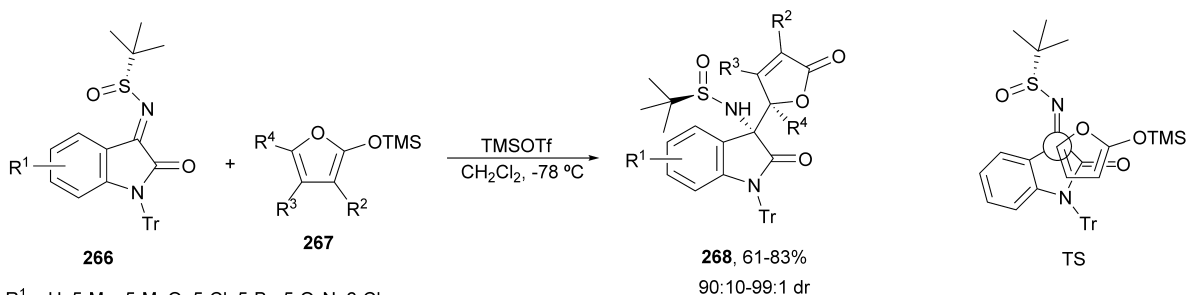
R⁴ = Ph, 4-MeOC₆H₄, BnO(CH₂)₂, *n*-C₆H₁₃, *n*-C₁₂H₂₅

Scheme 112. Reaction of imines **16** with silyl enol ethers **262**.



Ar = Ph, 4-O₂NC₆H₄, 4-MeO₂CC₆H₄, 4-F₃CC₆H₄, 3,5-(F₃C)₂C₆H₃, 4-MeC₆H₄, 4-MeOC₆H₄, 2-HSC₆H₄, 1-naphthyl

Scheme 113. Reaction of imines **16** with silyl enol ether **264**.



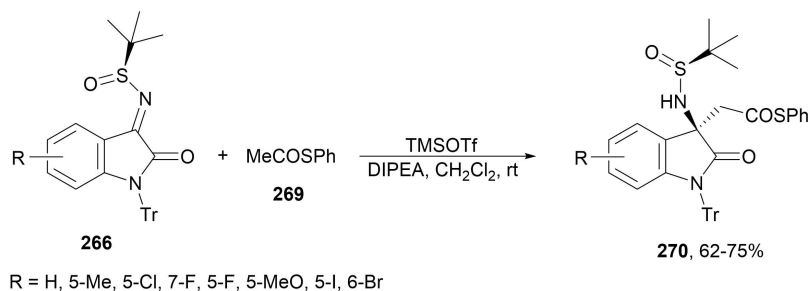
R¹ = H, 5-Me, 5-MeO, 5-Cl, 5-Br, 5-O₂N, 6-Cl

R² = H, Me

R³ = H, Me

R⁴ = H, Me, Et, CH₂=CHCH₂

Scheme 114. Reaction of imines **266** with furans **267**.



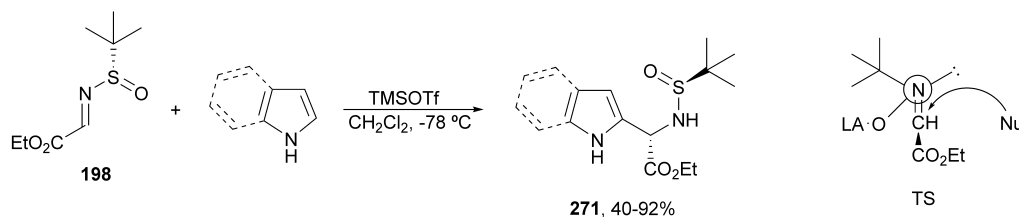
R = H, 5-Me, 5-Cl, 7-F, 5-F, 5-MeO, 5-I, 6-Br

Scheme 115. Reaction of imines **266** with enol ethers derived from **269**.

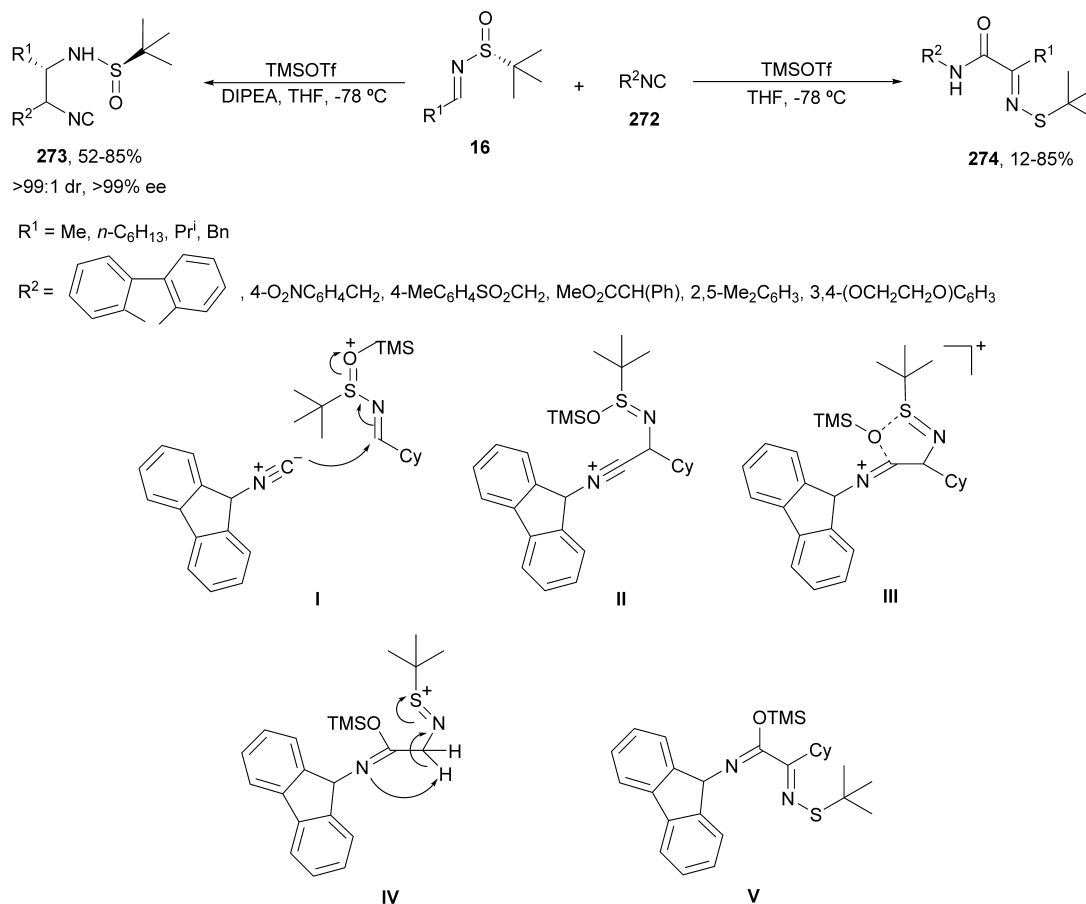
(Scheme 86).^[109] Transition state TS would explain the observed stereochemistry.

7. Organo Aluminum Compounds

In situ generated alkynyl aluminum compounds **195** were allowed to react with imines **16** to give the corresponding products **196** (Scheme 87).^[110]



Scheme 116. Lewis acid promoted reaction of imines **198** with pyrrole or indole.



Scheme 117. Reactions of imines **16** with isocyanides **272**.

8. Organoindium Compounds

The allylation^[111] of *N-tert*-butylsulfinyl imines in the presence of In metal^[112] is an interesting methodology, because after introducing the allyl group in the imine,^[113] it can be converted in other functional groups by simple transformations.

Apart from the general procedure to prepare the starting imines from the corresponding carbonyl compounds and

commercially available chiral *tert*-butylsulfinamides under thermal^[2] or MW^[3] activation, other methods starting from epoxides,^[114] or allylic alcohols and aryl iodides,^[115] are complementary, but not so general.

The allylation normally proceeded^[113] by treatment of the starting imine **16** with allyl bromide and In metal at 60 °C, and the stereochemistry of product **197** can be explained considering the transition state TS shown in Scheme 88. This methodology has been applied to allylic alcohols under Pd-

catalysis,^[116] and dieny bromides,^[117] as well as under aqueous conditions.^[118]

8.1. Allylation Reactions

The reaction depicted in Scheme 88 was also applied to ketimines giving the corresponding compounds with the nitrogen atom being in a quaternary center with good yields (44–92 %) and diastereoselectivities (59–41–>98:2 dr).^[119]

Imines **198** derived from α -keto and α -alkoxycarbonyl compounds were allylated in the presence of In metal in THF at room temperature to yield the expected products **199** resulting from an addition to the C=N bond (Scheme 89).^[120]

When the starting imine bears a ω -chloroalkyl substituent (**200**), the allylation gave compounds **201**, which by deprotection-cyclization yielded products **202**. Final treatment with trifluoroacetic acid (TFA) and Grubbs-II afforded the expected bicycles **203** (Scheme 90).^[121]

The allylation of imine **204** with allyl bromide either in THF or under aqueous conditions yielded, after treatment with LDA and Boc₂O, the corresponding lactam **205** (Scheme 91).^[122] This compound after several steps afforded epohelmin A.

For imines **16**, the allylation with the fluoroalkyl chloride **206** in the presence of In under standard conditions yielded the corresponding products **207** (Scheme 92).^[123] The reaction was also studied for one ketimine with similar results. Transition state TS was proposed to explain the stereochemistry observed in products **207**.

Once imines **16** were allylated under standard conditions, the corresponding products **197** were oxidized with MCPBA, so sulfonamide epoxides **208** were obtained as a *ca.* 1:1 diastereomeric mixture. This mixture was resolved with Jacobsen catalyst **209** to give a mixture of a diol **210**, and an epoxide **208** (Scheme 93). The diol was transformed into the same epoxide by known methodologies. Final epoxides **208** were easily transformed into azetidines and pyrrolidines by treatment with K₂CO₃ in the absence or presence of KI, through a 4-*exo-tet* or 5-*exo-tet* cyclization process, respectively.^[124]

An interesting improvement of the allylation process consists in the *in situ* generation of the starting imines, so in a one-pot reaction, the expected chiral protected amines were obtained (46–92 %, 79:21–>98:2 dr).^[125]

A simple application of compounds of type **197** was their transformation into 2,6-disubstituted piperidines **212**, *via* the α,β -unsaturated ketone **211**, by conventional methodologies (Scheme 94).^[126] Compound **212** belong to the family of natural alkaloids isosolenopsins.

The allyl bromide can bear an ester^[127] or a diester group.^[128] In the first case, after reaction of the allyl ester **213** with imines **16**, the corresponding amino esters **214** were

isolated (Scheme 95).^[127] Acid-base treatment of compounds **214** gave the expected lactams **215** with excellent yields.

In the case of aromatic imines **16**, their allylation with 1,3-dibromopropene **157** and In, under aqueous conditions, gave the corresponding sulfonamides **216**, which under basic conditions afforded aziridines **217** in a diastereoselective manner (Scheme 96).^[129]

When the starting imines **16** contains a bromine atom at the *ortho*-position, the allylation under standard conditions yielded the corresponding products **218** (Scheme 97).^[130] Compounds **218** underwent cyclization under Pd-catalysis, to give the expected methylenic bicycles **219**.

The same type of starting imines **16** shown in Scheme 97 reacted with the ester **213**, to give the corresponding products **220**, which by acidic-basic treatment gave compounds **221**, that by cycloaddition under CuI-catalysis yielded tricyclic derivatives **222** (Scheme 98).^[131] The allylation of imines **16**, followed by desulfonylation, *N*-acryloylation, and final ring-closing metathesis, afforded the corresponding tricyclic compounds.

An interesting molecule in order to synthesize several alkaloids is chiral 2-allylpiperidine **226**, easily prepared from 5-bromopentanal (**223**), and *tert*-butylsulfonamide, followed by allylation. The intermediate **224** was treated *in situ* with KHMDS in THF, yielding the protected piperidine **225**, which by final desulfonylation gave the expected compound **226** (Scheme 99).^[132] This compound was easily transformed into several alkaloids, such as (+)-coniine, (–)-pelletierine, (+)-allosedrinine, (+)-lasubine II, and cermizines, using conventional methodologies. For instance, the preparation of (–)-pelletierine took place by changing the protection by Boc (**227**), followed by a Wacker oxidation, and final deprotection.

Another application of amine **225** was the preparation of tetraoponerine T3. After change in the protection (to **228**), an oxidation gave compound **229**, which suffered a new allylation (through the corresponding imine), giving compound **230**. This compound was deprotected and hydrogenated to give intermediate **231**, which by *in situ* reaction with 4-bromobutanal gave the final alkaloid (Scheme 100).^[133] Following the same strategy, the whole family of tetraoponerines has been synthesized.^[133,134]

The synthesis of (–)-aphanorphine starts from homoallylic amine **232**, which was oxidized to compound **233** as a 1:1 diastereomer mixture. Change in the protection to **234**, followed by Friedel-Crafts reaction, afforded compound **235**, which by final *N*-methylation to **236**, and *O*-demethylation, gave the desired alkaloid (Scheme 101).^[135]

The allylation of the corresponding imines was the key step in the synthesis of the alkaloids (–)-tylophorine^[136] and 7-methoxycryptopleurine.^[137] On the other hand, Pd-promoted oxidation of compound **197** with benzoquinone (BQ) in

AcOH/DMSO produced the corresponding γ -acetoxy derivatives.^[138]

8.2. Propargylation Reactions

The In-promoted propargylation of imines **16** was carried out in THF under ultrasound activation, using a silylated propargyl bromide **237**, to give products **238** (Scheme 102). Compounds **238** can be orthogonally deprotected: with HCl/THF the primary amine is liberated, whereas with K_2CO_3 in THF/MeOH, the C–Si bond was cleaved, giving the terminal alkyne.^[139] Transition state TSI could explain the regiochemistry obtained, where no allenyl products were observed. On the other hand, transition state TSII was able to explain the stereochemistry observed in the final product **238**.

Compounds **239** come from the treatment of **238** with K_2CO_3 , as it was commented above. These compounds reacted with 2-iodophenol or 2-iodoaniline (**240**) to give mainly the corresponding furans **241**, or indoles **243**, respectively. In the second case, an additional step, heating at 100 °C intermediates **242**, was necessary to get the final heterocycles (Scheme 103).^[140]

The *N*-allylation of compounds **239** followed by ring-closing metathesis led to the formation of the corresponding bicyclic dienes, able to perform Diels-Alder reactions.^[141] The corresponding *N*-propargylation of **239** yielded diynes **244**, which after oxidation to compounds **245**, reacted with an acetylene in the presence of a Rh catalyst, to give the corresponding tetrahydroisoquinolines **246** (Scheme 104).^[142]

An intramolecular propargylation occurred when compound **247** was treated with In and TFA in THF/MeOH, giving the corresponding allenylated products **248** (Scheme 105).^[143] Favored transition state depicted in Scheme 105 would explain the region and stereochemistry of the process.

From the results included in this Section, we conclude that the allylation and propargylation of *N*-*tert*-butylsulfinyl imines in the presence of In metal is a versatile procedure to prepare chiral homoallylic, or homopropargylic amines. These compounds are easily transformed into interesting compounds, mainly in the field of alkaloids.

9. Organosilicon Compounds

In this Section, the reaction of imines with organosilicon compounds, silicon enolates, and reactions with silicon as Lewis acid will be considered.

9.1. Reaction with Organosilicon Compounds

Imines **16** react with organosilicon compounds **249** in the presence of TMSOK to give the corresponding products **250** (Scheme 106).^[144] The process is rather general, aliphatic, aromatic and heteroaromatic organosilicon compounds being used with good results in terms of yields and diastereoselectivities.

The reaction of imines **16** with the base LDA and an organosilicon compound $TBSCHCl_2$, or $TMSCHCl_2$ (**251**) in the presence of an aryl cyanide **252**, led to the formation of aziridines **253**, which stereochemistry depends on the starting silicon compound (Scheme 107).^[145] For $Si=TBS$, the corresponding aziridines **253a** were obtained, whereas for $Si=TMS$, the other diastereomers **253b** were the reaction products. Intermediates **I**→**IV** are probably involved in the reaction, the step **II**→**III** being a [1,3]-aza-Brook rearrangement. Concerning the observed stereochemistry, an open transition state TSI was proposed for $Si=TBS$, while a cyclic TSII was probably involved in the case of $Si=TMS$, so steric effects are responsible of the obtained results.

The reaction of $PhMe_2SiLi$ (**254**) with imines **198** derived from a α -ketoester, and an aldehyde, in the presence of a mixture of THF and HMPA at low temperature, yielded compounds **255**, which in several cases have been oxidized to the corresponding sulfonamides with MCPBA (Scheme 108).^[146] An open transition state TS was formulated in order to explain the observed stereochemistry in **255**.

The reaction of imines **16** with the same organolithium compound **254**, and $Ar^2COCON(CH_2)_5$ (**256**) led also to the corresponding products **257** with good yields and high diastereoselectivities (Scheme 109).^[147] Transition state TS would explain the observed stereochemistry in compounds **257**.

Finally, $TMSCN$ added to imines **258** in the presence of Na_2CO_3 in THF, to yield compound **259**, which was used as intermediate in the synthesis of hydroxypipelicolic acid (Scheme 110).^[148]

9.2. Reaction with Silyl Enol Ethers

The reaction of silyl enol ethers **260** derived from α,β -unsaturated ketones with imines **16** in the presence of $TMSOTf$ gave the corresponding compounds **261** (Scheme 111).^[149] Some of these compounds were easily transformed into lasubine alkaloids. In addition, an open transition state can explain the observed stereochemistry.

Silyl enol ethers **262** derived from ynones^[150] reacted with imines **16** under the conditions depicted in Scheme 112 to yield the corresponding products **263** (Scheme 112).^[151] Some of these compounds were easily transformed into the phytosphingosine alkaloids family.

When imines **16** react with trimethylsilyl ketene acetal **264** in the presence of TBAT gave the corresponding protected aminoalcohols, which *in situ* were desulfinylated to yield the expected amino acids **265**, isolated as their hydrochlorides (Scheme 113).^[152] Some of the obtained products are cyclic amino acids isolated with good yields (49–99%) and diastereoselectivities (90:10–94:6 dr).

The reaction of imines **266** derived from isatine with TMS-derived from furan **267** in the presence of TMSOTf at low temperature yielded the corresponding compounds **268** (Scheme 114).^[153] The transition state TS was formulated in order to explain the observed stereochemistry.

Finally, the same type of imines **266** reacted with the silyl enolates of *S*-phenyl thioacetate (**269**) in an asymmetric Mannich reaction to yield the corresponding products **270** (Scheme 115).^[154]

9.3. Silicon as Lewis acid

The reaction of pyrrole or indole with imines **198**, in the presence of TMSOTf at low temperature, gave sulfinyl amino esters **271** (Scheme 116).^[155] The open chair model TS shown in Scheme 116 can explain the observed stereochemistry in products **271**.

Imines **16** react with isocyanides **272** in the presence of TMSOTf to give different products, depending on the presence or absence of DIPEA. In the presence of this base, compounds **273** were isolated with good yields and excellent diastereoselectivities (Scheme 117).^[156] However, in the absence of the base, compounds **274** were the reaction products.^[157] DFT calculations support the mechanism depicted in Scheme 117, involving intermediates **I–V**.

10. Conclusion

Some conclusions are included at the end of the most important Sections. As general conclusion, we believe that the addition of organometallic reagents to chiral *N*-*tert*-butylsulfinyl imine is an excellent method to prepare chiral protected amines, which can be deprotected easily under acidic conditions to yield the corresponding primary amines. This chemistry has been applied to the synthesis of interesting nitrogenated compounds, especially in the field of alkaloids. In the next future we believe that the use of other nucleophiles different than organometallic reagents derived from main-group metals will open a new window to prepare interesting chiral nitrogen-containing compounds from natural or unnatural origin.

Abbreviations

Ac	Acetyl
aq	Aqueous
Ar	Aryl
Bn	Benzyl
Boc	<i>tert</i> -Butoxycarbonyl
Bpin	Pinacolato boron
BQ	Benzoquinone
<i>ca.</i>	Circa
cat.	Catalyst, catalytic amount
Cbz	Benzyloxycarbonyl
Cy	Cyclohexyl
DCE	1,2-Dichloroethane
de	Diastereomeric excess
DFT	Density functional theory
DIPEA	Diisopropylethylamine
DMF	<i>N,N</i> -Dimethylformamide
DMSO	Dimethylsulfoxide
dppe	1,2-Bis(diphenylphosphino)ethane
dr	Diastereomeric ratio
DTBS	Di- <i>tert</i> -butylsilyl bis(trifluoromethanesulfonate)
ee	Enantiomeric excess
HMPA	Hexamethylphosphoramide
LAH	Lithium aluminum hydride
LB	Lewis base
LDA	Lithium diisopropyl amine
LiHMDS	Lithium hexamethyldisilazane
LiTMP	Lithium 2,2,6,6-tetramethylpiperidine
MCPBA	<i>meta</i> -Chloroperbenzoic acid
Moc	Methoxycarbonyl
MS	Molecular sieves
MW	Microwaves
NaHMDS	Sodium hexamethyldisilazane
Np	6-Nitropiperonylmethyl
N-Selectride	(2-Butyl) ₃ BHNa
P4- <i>t</i> -Bu	1- <i>tert</i> -Butyl-4,4,4-tris(dimethylamino)-2,2-bis[tris(dimethylamino)phosphoranylideneimino]-2λ ⁵ ,4 λ ⁵ -catenadi(phosphacene)
PDC	Pyridinium dichromate
PMB	4-Methoxybenzyl
PMP	4-Methoxyphenyl
Py	Pyridyl
RCM	Ring closing metathesis
rt	Room temperature
s	Solvent
TBAT	Tetrabutyl ammonium triphenyldifluorosilicate
TBDPS	<i>tert</i> -Butyl diphenyl silyl
TBS	<i>tert</i> -Butyl dimethyl silyl
Tf	Triflate, triphenylmethylsulfonyl
TFA	Trifluoroacetic acid
THF	Tetrahydrofuran

THP	Tetrahydropyranyl
TMEDA	Tetramethylethylenediamine
TMS	Trimethylsilyl
TMSOI	Trimethylsulfonium iodide
Tr	Trityl, triphenylmethylsulfonyl
Ts	Tosyl, 4-methylphenyl
TS	Transition state

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References

- [1] J. Ramirez Hernandez, F. Chemla, F. Ferreira, O. Kackowski, J. Oble, A. Perez-Luna, G. Poli, *Chimia* **2016**, *70*, 84–92.
- [2] G. Liu, D. A. Cogan, T. D. Owens, T. P. Tang, J. A. Ellman, *J. Org. Chem.* **1999**, *64*, 1278–1284.
- [3] J. F. Collados, E. Toledano, D. Guijarro, M. Yus, *J. Org. Chem.* **2012**, *77*, 5744–5750.
- [4] a) D. Guijarro, O. Pablo, M. Yus, *Org. Synth.* **2013**, *90*, 338–349; b) M. Yus, *Org. Synth.* **2019**, *96*, 232–244.
- [5] F. Foubelo, M. Yus, *Chem. Rec.* **2015**, *15*, 907–924.
- [6] a) D. Guijarro, O. Pablo, M. Yus, *Tetrahedron Lett.* **2009**, *50*, 5386–5388; b) D. Guijarro, O. Pablo, M. Yus, *J. Org. Chem.* **2010**, *75*, 5265–5270.
- [7] D. Guijarro, O. Pablo, M. Yus, *Tetrahedron Lett.* **2011**, *52*, 789–791.
- [8] O. Pablo, D. Guijarro, G. Kovács, A. Lledós, G. Ujaque, M. Yus, *Chem. Eur. J.* **2012**, *18*, 1969–1983.
- [9] O. Pablo, D. Guijarro, M. Yus, *Eur. J. Org. Chem.* **2014**, 7034–7038.
- [10] O. Pablo, D. Guijarro, M. Yus, *J. Org. Chem.* **2013**, *78*, 3647–3654.
- [11] D. Guijarro, O. Pablo, M. Yus, *J. Org. Chem.* **2013**, *78*, 9181–9189.
- [12] T. Liu, L.-y. Chen, Z. Sun, *J. Org. Chem.* **2015**, *80*, 11441–11446.
- [13] M. Liu, Y. Xie, J. Li, H. Pan, H. Tian, Y. Shi, *J. Org. Chem.* **2014**, *79*, 8417–8421.
- [14] P. Liu, Z.-J. Liu, F. Wu, *Adv. Synth. Catal.* **2015**, *357*, 818–822.
- [15] H. Mei, Y. Xiong, C. Xie, V. A. Soloshonok, J. Han, Y. Pan, *Org. Biomol. Chem.* **2014**, *12*, 2108–2113.
- [16] C. Xie, L. Wu, H. Mei, V. A. Soloshonok, J.-H. Han, Y. Pan, *Org. Biomol. Chem.*, **2014**, *12*, 7836–7843.
- [17] L. R. Reddy, S. Kotturi, Y. Waman, C. Patel, M. Danidharia, R. Shenoy, *J. Org. Chem.* **2018**, *83*, 6573–6579.
- [18] W. J. Morris, K. K. Muppalla, C. Cowden, R. G. Ball, *J. Org. Chem.* **2013**, *78*, 706–710.
- [19] M. Sanchez Roselló, C. del Pozo, S. Fustero, *Synthesis* **2016**, *48*, 2553–2571.
- [20] I. Chogii, P. Das, M. D. Delost, M. N. Crawford, J. T. Njardarson, *Org. Lett.* **2018**, *20*, 4942–4945.
- [21] M. B. Uphade, A. Amaranadha Reddy, S. P. Khandare, K. R. Prasad, *Org. Lett.* **2019**, *21*, 9109–9113.
- [22] Y. Chen, F. W. Goldberg, J. Xiong, S. Wang, *Synthesis* **2015**, *47*, 679–691.
- [23] L. Rajender Reddy, S. Kottri, Y. Waman, C. Patel, A. Patwa, R. Shenoy, *Chem. Commun.* **2018**, *54*, 7007–7009.
- [24] R. Chinchilla, C. Nájera, M. Yus, *Tetrahedron* **2005**, *61*, 3139–3176.
- [25] M. Yus, *Pure Appl. Chem.* **2003**, *75*, 1453–1475.
- [26] J. Almena, F. Foubelo, M. Yus, *Tetrahedron* **1995**, *51*, 3351–3364.
- [27] D. García, B. Moreno, T. Soler, F. Foubelo, M. Yus, *Tetrahedron Lett.* **2009**, *50*, 4710–4713.
- [28] S. Jordan, S. A. Starks, M. F. Whatley, M. Turlington, *Org. Lett.* **2015**, *17*, 4842–4845.
- [29] C. Verrier, S. Carret, J.-F. Poisson, *Org. Lett.* **2012**, *14*, 5122–5125.
- [30] P. Cividino, C. Verrier, C. Philouze, S. Carret, J.-F. Poisson, *Adv. Synth. Catal.* **2019**, *361*, 1236–1240.
- [31] X.-N. Wang, R. P. Hsung, R. Qi, S. K. Fox, M.-C. Lv, *Org. Lett.* **2013**, *10*, 2514–2517.
- [32] D. N. Penk, N. A. Robinson, H. M. Hill, M. Turlington, *Tetrahedron Lett.* **2017**, *58*, 470–473.
- [33] Y. Dai, C. Xie, H. Mei, J. Han, V. A. Soloshonok, *Tetrahedron* **2015**, *71*, 9550–9556.
- [34] J.-b. Zhao, X. Pen, B.-q. Zheng, J. Ji, Z.-b. Qiu, Y. Li, *Tetrahedron Lett.* **2018**, *59*, 2091–2094.
- [35] J. Guernon, L. Marcin, M. Higgins, F. Yang, J. Shi, L. Snyder, L. A. Thompson, *Tetrahedron Lett.* **2014**, *55*, 2134–2137.
- [36] a) X. Li, Y. Li, H. Shang, *Org. Biomol. Chem.* **2016**, *14*, 6457–6462; b) H. Shang, Y. Li, X. Li, X. Ren, *J. Org. Chem.* **2015**, *80*, 8739–8747.
- [37] N. Shibata, T. Nishimine, N. Shibata, E. Tokunaga, K. Kawada, T. Kagawa, J. L. Azeña, A. Sorochinsky, V. Soloshonok, *Org. Biomol. Chem.* **2014**, *12*, 1454–1462.
- [38] T. Moragas, I. Churcher, W. Lewis, R. A. Stockman, *Org. Lett.* **2014**, *16*, 6290–6293.
- [39] F. A. Davies, H. Liu, G. V. Reddy, *Tetrahedron Lett.* **1996**, *37*, 5473–5476.
- [40] L. R. Reddy, Y. Waman, P. Kallure, K. S. Nalivela, Z. Begum, T. Divay, S. Kottury, *Chem. Commun.* **2019**, *55*, 5968–5070.
- [41] X. Chen, Y. Li, J. Zhao, B. Cheng, Q. Lu, X. Ren, *Adv. Synth. Catal.* **2017**, *359*, 3057–3062.
- [42] I. Chogii, J. T. Njardarson, *Angew. Chem. Int. Ed.* **2015**, *54*, 13706–13710.
- [43] H. Wang, P. Tang, Q. Zhou, D. Zhang, Z. Chen, H. Huang, Y. Qin, *J. Org. Chem.* **2015**, *80*, 2494–2502.
- [44] Y.-X. Zhang, L.-Y. Chen, B.-G. Wei, *Tetrahedron Lett.* **2020**, *61*, 151460.

- [45] H. Zhang, Y. Li, F. Xu, X. F. Ren, L. Wang, P. Zhou, Z. Sun, *Synlett* **2012**, 23, 2485–2490.
- [46] Y. Li, D. Li, T. Zheng, H. Li, X. Ren, *Chem. Eur. J.* **2014**, 20, 14986–14990.
- [47] X. Li, D. G. Hall, *Angew. Chem. Int. Ed.* **2018**, 57, 10304–10308.
- [48] Y. Zhang, H. Tian, Z. Luo, X. Liu, Y. Zhao, W. Chen, H. Zhang, *Tetrahedron Lett.* **2018**, 59, 3357–3360.
- [49] Z.-F. Li, Y. Yao, Y.-J. Xu, C.-D. Lu, *J. Org. Chem.* **2019**, 84, 7207–7219.
- [50] J. Feng, P.-J. Ma, Y.-M. Zeng, Y.-J. Xu, C.-D. Lu, *Chem. Commun.* **2018**, 54, 2882–2885.
- [51] W. Huang, H. Liu, C.-D. Lu, Y.-J. Xu, *Chem. Commun.* **2016**, 52, 13592–13595.
- [52] P. Chen, X. Su, W. Zhou, Y. Xiao, J. Zhang, *Tetrahedron* **2016**, 72, 2700–2706.
- [53] J. T. Reeves, Z. Tan, M. A. Herbage, Z. S. Han, M. A. Marsini, Z. Li, G. Li, Y. Xu, K. R. Fandrick, N. C. Gonella, S. Campbell, S. Ma, N. Grinberg, N. Lee, B. Z. Lu, C. H. Senanayake, *J. Am. Chem. Soc.* **2013**, 135, 5565–5568.
- [54] a) S. Zhao, R. B. Andrade, *J. Am. Chem. Soc.* **2013**, 135, 13334–13337; b) S. Zhao, R. B. Andrade, *J. Org. Chem.* **2017**, 82, 521–531.
- [55] L. Wu, C. Xie, H. Wei, Y. Sai, J. Han, V. A. Soloshonok, Y. Pan, *J. Org. Chem.* **2015**, 80, 3187–3194.
- [56] W. Zhang, X. Wang, B. Zhu, D. Zhu, J. Han, A. Wzorek, A. Sato, V. A. Soloshonok, J. Zhou, Y. Pan, *Adv. Synth. Catal.* **2017**, 359, 4267–4273.
- [57] P. J. Ma, H. Liu, C.-D. Lu, Y. J. Xu, *Org. Lett.* **2017**, 19, 670–673.
- [58] P. J. Ma, H. Liu, Y.-J. Xu, H. A. Aisa, C.-D. Lu, *Org. Lett.* **2018**, 20, 1236–1239.
- [59] S. Qin, S. Liu, Y. Cao, J. Li, C. Chong, T. Liu, Y. Luo, J. Hu, S. Jiang, H. Zhou, G. Yang, C. Yang, *Org. Lett.* **2018**, 20, 1350–1354.
- [60] Y. Li, Y. Ma, Z. Lu, L. Wang, X. Ren, Z. Sun, *Tetrahedron Lett.* **2012**, 53, 4711–4714.
- [61] Y. Yang, Y. Huang, F.-L. Qing, *Tetrahedron Lett.* **2013**, 54, 3826–3830.
- [62] X. Liu, Y. Yang, X. Huang, X.-H. Xu, F.-L. Qing, *Synlett* **2015**, 26, 67–72.
- [63] C.-T. Li, H. Liu, Y.-J. Xu, C. P. Lu, *J. Org. Chem.* **2017**, 82, 11253–11261.
- [64] N. Punna, T. Saito, M. Kosobokov, E. Tokunaga, Y. Sumii, N. Shibata, *Chem. Commun.* **2018**, 54, 4294–4297.
- [65] C.-T. Li, H. Liu, Y. Tao, C.-D. Lu, *Org. Lett.* **2019**, 21, 8383–8388.
- [66] C. Xie, H. Mei, L. Wu, V. A. Soloshonok, J. Han, Y. Pan, *Eur. J. Org. Chem.* **2014**, 1445–1451.
- [67] J. A. Sirvent, F. Foubelo, M. Yus, *Heterocycles* **2014**, 88, 1163–1174.
- [68] C.-M. Si, Z.-Y. Mao, H.-Q. Dong, Z.-T. Du, B.-G. Wei, G.-Q. Lin, *J. Org. Chem.* **2015**, 80, 5824–5833.
- [69] M. Hennum, H. Fliegl, L.-L. Gundersen, O. Eisenstein, *J. Org. Chem.* **2014**, 79, 2514–2521.
- [70] F. Grellepois, A. B. Jamaa, A. Gassama, *Eur. J. Org. Chem.* **2013**, 6694–6701.
- [71] C. Cocaud, C. Nicolas, A. Bayle, T. Poisson, X. Pannecouche, O. R. Martin, *Eur. J. Org. Chem.* **2015**, 4330–4334.
- [72] N. S. S. Reddy, B. J. M. Reddy, B. V. Reddy, *Tetrahedron Lett.* **2013**, 54, 4228–4231.
- [73] D. Chen, M.-H. Xu, *J. Org. Chem.* **2014**, 79, 7746–7751.
- [74] C. Pierry, D. Cahard, S. Couve-Bonnaire, X. Pannecouche, *Org. Biomol. Chem.* **2011**, 9, 2378–2386.
- [75] J.-L. Jiang, M. Yao, C.-D. Lu, *Org. Lett.* **2014**, 16, 318–321.
- [76] S. G. Davies, A. M. Fletcher, P. M. Roberts, R. S. Shah, A. L. Thompson, J. E. Thomson, *Org. Lett.* **2014**, 16, 1354–1357.
- [77] a) C.-M. Si, W. Huang, Z.-T. Du, B.-G. Wei, G.-Q. Lin, *Org. Lett.* **2014**, 16, 4328–4331; b) Y.-W. Liu, Z.-W. Mao, R.-J. Ma, J.-H. Yan, C.-M. Si, B.-G. Wei, *Tetrahedron* **2017**, 73, 2100–2108.
- [78] Q.-R. Zhou, X.-Y. Wei, Y.-Q. Li, D. Huang, B.-G. Wei, *Tetrahedron* **2014**, 70, 4799–4808.
- [79] C. M. Si, L.-P. Shao, Z. Y. Mao, W. Zhou, B.-G. Wei, *Org. Biomol. Chem.* **2017**, 15, 649–661.
- [80] S. Pindi, J. Wu, G. Li, *J. Org. Chem.* **2013**, 78, 4006–4012.
- [81] a) N. S. S. Reddy, B. V. S. Reddy, *Tetrahedron Lett.* **2014**, 55, 3143–3145; b) N. S. S. Reddy, R. A. Babu, B. V. S. Reddy, *Synthesis* **2016**, 48, 1079–1086.
- [82] S. Chowdhury, J. Mao, J. Martynow, H. Zhao, E. Duffy, Y. Wu, V. Thakur, G. Sirasani, Y. Tang, F. Collin, S. Sinishtaj, A. Bhatlacharjee, *Tetrahedron Lett.* **2019**, 60, 371–374.
- [83] B. Wang, P. P. Zhang, *Tetrahedron Lett.* **2012**, 53, 119–122.
- [84] K. Fjelbye, N. Svenstrup, A. Püschl, *Synthesis* **2015**, 47, 3231–3240.
- [85] J. A. Mendes, P. Merino, T. Soler, E. J. Salustiano, P. R. R. Costa, M. Yus, F. Foubelo, C. D. Buarque, *J. Org. Chem.* **2019**, 84, 2219–2233.
- [86] H. T. Cao, T. Roisnel, A. Valleix, R. Greé, *Eur. J. Org. Chem.* **2011**, 3430–3436.
- [87] A. Adamkiewicz, J. Mlynarski, *Eur. J. Org. Chem.* **2016**, 1060–1065.
- [88] W.-J. Liu, Y.-H. Zhao, X.-W. Sun, *Tetrahedron Lett.* **2013**, 54, 3586–3590.
- [89] S. Fustero, L. Herrera, R. Lázaro, E. Rodriguez, M. A. Maestro, N. Mateu, P. Barrio, *Chem. Eur. J.* **2013**, 19, 11776–11785.
- [90] F. Grellepois, *J. Org. Chem.* **2013**, 78, 1127–1137.
- [91] L. Su, M.-H. Xu, *Synthesis* **2016**, 48, 2595–2602.
- [92] Y.-Y. Peng, P.-Liu, Z.-J. Liu, J.-T. Liu, H.-F. Mao, Y.-L. Yao, *Tetrahedron* **2018**, 74, 3074–3080.
- [93] R. Almansa, D. Guijarro, M. Yus, *Tetrahedron: Asymmetry* **2008**, 19, 2484–2491.
- [94] R. Almansa, D. Guijarro, M. Yus, *Tetrahedron Lett.* **2009**, 50, 3198–3201.
- [95] a) R. Almansa, D. Guijarro, M. Yus, *Tetrahedron Lett.* **2009**, 50, 4188–4190; b) R. Almansa, J. F. Collados, D. Guijarro, M. Yus, *Tetrahedron: Asymmetry* **2010**, 21, 1421–1431.
- [96] T. Guo, B.-H. Yuan, W.-J. Liu, *Org. Biomol. Chem.* **2018**, 16, 57–61.
- [97] F. Grellepois, A. B. Jamaa, N. S. Rosa, *Org. Biomol. Chem.* **2017**, 15, 9696–9709.
- [98] L. Liu, J. Liu, *Chin. J. Chem.* **2011**, 29, 2722–2726.

- [99] S. Fustero, R. Lázaro, E. Rodríguez, N. Mateu, P. Barrio, *Org. Lett.* **2013**, *15*, 3770–3773.
- [100] N. Y. Kuznetsov, R. M. Tikhov, I. A. Godovikov, M. G. Medvedev, K. A. Lyssenko, E. I. Burtseva, E. S. Kirillova, Y. N. Bubnov, *Org. Biomol. Chem.* **2017**, *15*, 3152–3157.
- [101] A. Shen, Z.-T. He, H.-J. Yu, Y. Fukui, P. Tiang, C.-Q. Lin, *Synthesis* **2013**, 1649–1656.
- [102] D. Chen, M.-H. Xu, *Chem. Commun.* **2013**, *49*, 1327–1329.
- [103] W. Yan, D. Wang, J. Feng, P. Li, R. Wang, *J. Org. Chem.* **2012**, *77*, 3311–3317.
- [104] a) J. C. González-Gómez, F. Foubelo, M. Yus, *Tetrahedron Lett.* **2008**, *49*, 2343–2347; b) J. C. González-Gómez, F. Foubelo, M. Yus, *J. Org. Chem.* **2009**, *74*, 2547–2553; c) J. C. González-Gómez, F. Foubelo, M. Yus, *Synthesis* **2009**, 2083–2088.
- [105] W. Huang, J.-L. Ye, W. Wang, H.-Q. Dong, B. G. Wei, *J. Org. Chem.* **2013**, *78*, 11229–11237.
- [106] L. Wu, C. Xie, H. Mei, W. A. Soloshonok, J. Han, Y. Pan, *J. Org. Chem.* **2014**, *79*, 7677–7681.
- [107] S. Sugiyama, S. Imani, K. Ishii, *Tetrahedron: Asymmetry* **2013**, *24*, 1069–1074.
- [108] C. Xie, L. Zhang, H. Mei, R. Pajkert, M. Ponomarenko, Y. Pan, G.-V. Röschenthaler, V. A. Soloshonok, J. Han, *Chem. Eur. J.* **2016**, *22*, 7036–7040.
- [109] Y. Li, M.-H. Xu, *Org. Lett.* **2012**, *14*, 2062–2065.
- [110] C. Verrier, S. Carret, J.-F. Poisson, *Org. Biomol. Chem.* **2014**, *12*, 1875–1878.
- [111] a) M. Yus, J. C. González-Gómez, F. Foubelo, *Chem. Rev.* **2011**, *101*, 7774–7854; b) M. Yus, J. C. González-Gómez, F. Foubelo, *Chem. Rev.* **2013**, *113*, 5595–5698.
- [112] a) J. C. González-Gómez, F. Foubelo, M. Yus, *Org. Synth.* **2012**, *89*, 88–97; b) M. Yus, *Org. Synth.* **2019**, *96*, 179–194; c) F. Foubelo, M. Yus, *Eur. J. Org. Chem.* **2014**, 485–491; d) F. Foubelo, M. Yus, *Russ. Chem. Bull. Int. Ed.* **2016**, *65*, 1667–1686; e) F. Foubelo, M. Yus, *Chem. Today* **2016**, *34*, 45–49.
- [113] F. Foubelo, M. Yus, *Tetrahedron: Asymmetry* **2004**, *15*, 3823–3825.
- [114] A. Lahosa, F. Foubelo, M. Yus, *Eur. J. Org. Chem.* **2016**, 4067–4076.
- [115] S. Ikhlef, C. Behloul, A. Lahosa, F. Foubelo, M. Yus, *Eur. J. Org. Chem.* **2018**, 2609–2614.
- [116] O. Soares do Rego Barros, J. A. Sirvent, F. Foubelo, M. Yus, *Chem. Commun.* **2014**, *50*, 6898–6901.
- [117] a) Y. Nishigaichi, S. Tsuruta, K. Uenaga, T. Awamura, H. Iwamoto, A. Takuwa, *Tetrahedron Lett.* **2014**, *55*, 510–513; b) I. Bosque, E. Bagdatli, F. Foubelo, M. Yus, *J. Org. Chem.* **2014**, *79*, 1796–1804.
- [118] X.-W. Sun, M. Liu, M.-H. Xu, G.-Q. Lin, *Org. Lett.* **2008**, *10*, 1259–1262.
- [119] J. A. Sirvent, F. Foubelo, M. Yus, *Chem. Commun.* **2012**, *48*, 2543–2544.
- [120] a) E. Maciá, F. Foubelo, M. Yus, *Tetrahedron: Asymmetry* **2017**, *28*, 1407–1413; b) E. Maciá, F. Foubelo, M. Yus, *Tetrahedron* **2016**, *72*, 6001–6010.
- [121] T. J. Senter, M. L. Schulte, L. C. Konkol, T. E. Wadzinski, C. W. Lidsley, *Tetrahedron Lett.* **2013**, *54*, 1645–1648.
- [122] Y.-W. Liu, P. Han, W. Zhou, Z.-Y. Mao, C.-M. Si, B.-G. Wei, *Org. Biomol. Chem.* **2016**, *14*, 10714–10722.
- [123] G. Lemonnier, N. Van Hijfte, T. Poisson, S. Couve-Bonnaire, X. Pannecouche, *J. Org. Chem.* **2014**, *79*, 2916–2925.
- [124] M. Medjahdi, J. C. González-Gómez, F. Foubelo, M. Yus, *J. Org. Chem.* **2009**, *74*, 7859–7865.
- [125] J. C. González-Gómez, M. Medjahdi, F. Foubelo, M. Yus, *J. Org. Chem.* **2010**, *75*, 6308–6311.
- [126] a) J. C. González-Gómez, F. Foubelo, M. Yus, *Synlett* **2008**, 2777–2780; b) M. Medjahdi, J. C. González-Gómez, F. Foubelo, M. Yus, *Heterocycles* **2012**, *86*, 727–734.
- [127] a) H. K. Dema, F. Foubelo, M. Yus, *Heterocycles* **2010**, *80*, 125–131; b) H. K. Dema, F. Foubelo, M. Yus, *Heterocycles* **2011**, *82*, 1411–1421.
- [128] H. K. Dema, F. Foubelo, M. Yus, *Jordan J. Chem.* **2011**, *6*, 247–255.
- [129] E. Maciá, F. Foubelo, M. Yus, *Heterocycles* **2019**, *99*, 248–266.
- [130] J. A. Sirvent, F. Foubelo, M. Yus, *Eur. J. Org. Chem.* **2013**, 2461–2471.
- [131] J. A. Sirvent, F. Foubelo, M. Yus, *J. Org. Chem.* **2014**, *79*, 1356–1367.
- [132] I. Bosque, J. C. González-Gómez, F. Foubelo, M. Yus, *J. Org. Chem.* **2012**, *77*, 780–784.
- [133] I. Bosque, J. C. González-Gómez, A. Guijarro, F. Foubelo, M. Yus, *J. Org. Chem.* **2012**, *77*, 10340–10346.
- [134] I. Bosque, J. C. González-Gómez, M. I. Loza, J. Brea, *J. Org. Chem.* **2014**, *79*, 3982–3991.
- [135] M. Medjahdi, J. C. González-Gómez, F. Foubelo, M. Yus, *Eur. J. Org. Chem.* **2011**, 2230–2234.
- [136] a) C. Anton-Torrecillas, J. C. González-Gómez, *Org. Biomol. Chem.* **2014**, *12*, 7018–7025; b) C. Anton-Torrecillas, M. Loza, J. Brea, J. C. González-Gómez, *Org. Biomol. Chem.* **2016**, *14*, 2264–2271.
- [137] C. Anton-Torrecillas, I. Bosque, J. C. González-Gómez, M. Loza, J. Brea, *J. Org. Chem.* **2015**, *80*, 1284–1290.
- [138] A. Sirvent, T. Soler, F. Foubelo, M. Yus, *Chem. Commun.* **2017**, *53*, 2701–2704.
- [139] M. J. García-Muñoz, F. Zacconi, F. Foubelo, M. Yus, *Eur. J. Org. Chem.* **2013**, 1287–1295.
- [140] M. J. García-Muñoz, F. Foubelo, M. Yus, *J. Org. Chem.* **2016**, *81*, 10214–10226.
- [141] M. J. García-Muñoz, A. Sirvent, F. Foubelo, M. Yus, *An. Acad. Bras. Cienc.* **2018**, *90*, 1059–1072.
- [142] A. Sirvent, M. J. García-Muñoz, M. Yus, F. Foubelo, *Eur. J. Org. Chem.*, **2020**, 113–126.
- [143] L. Su, T.-S. Zhu, M.-H. Xu, *Org. Lett.* **2014**, *16*, 4118–4121.
- [144] a) M. Das, D. F. O'Shea, *Org. Lett.* **2015**, *17*, 1962–1965; b) M. Das, D. F. O'Shea, *Chem. Eur. J.* **2015**, *21*, 18717–18723.
- [145] Z.-A. Huang, H. Liu, C.-D. Lu, Y.-J. Xu, *Org. Lett.* **2015**, *17*, 4042–4045.
- [146] W. Huang, H. Liu, Y.-J. Xu, C.-D. Lu, *J. Org. Chem.* **2017**, *82*, 10748–10755.
- [147] Z. Sun, H. Liu, Y.-M. Zeng, C.-D. Lu, Y.-J. Xu, *Org. Lett.* **2016**, *18*, 620–623.
- [148] Z. Zhang, Z. Sun, *Synlett* **2020**, *31*, 355–358.

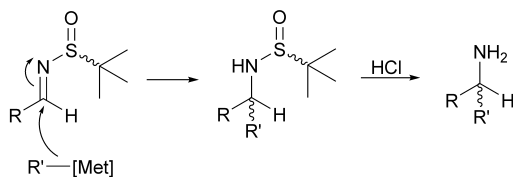
- [149] a) A. A. Reddy, P. O. Reddy, K. R. Prasad, *J. Org. Chem.* **2016**, *81*, 11363–11371; b) A. A. Reddy, K. R. Prasad, *J. Org. Chem.* **2017**, *82*, 13488–13499.
- [150] C. Nájera, L. K. Sydnes, M. Yus, *Chem. Rev.* **2019**, *119*, 11110–11246.
- [151] P. O. Reddy, A. A. Reddy, K. R. Prasad, *J. Org. Chem.* **2020**, *85*, 2743–2751.
- [152] M. Cantú-Reyes, I. Alvarado-Beltrán, R. Ballinas-Indili, C. Álvarez-Toledano, M. Hernández-Rodríguez, *Org. Biomol. Chem.* **2017**, *15*, 7705–7709.
- [153] V. U. Rao, A. P. Jadhav, D. Garad, R. P. Sing, *Org. Lett.* **2014**, *16*, 648–651.
- [154] S. Hajra, S. S. Bhosale, A. Hazra, *Org. Biomol. Chem.* **2017**, *15*, 9217–9225.
- [155] T. Andreassen, L.-K. Hansen, O. R. Gautun, *Eur. J. Org. Chem.* **2008**, 4871–4876.
- [156] G. V. Janssen, E. Vicente-García, W. Vogel, J. C. Slootweg, E. Ruitger, K. Lammertsma, R. V. A. Orru, *Eur. J. Org. Chem.* **2014**, 3762–3766.
- [157] G. V. Janssen, E. Janssen, C. M. L. Vande Velde, A. W. Ehlers, J. C. Slootweg, E. Ruitger, K. Lammertsma, R. V. A. Orru, *Org. Lett.* **2014**, *16*, 5116–5119.

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PERSONAL ACCOUNT



[Met] = Li, Na, K, Cs, Mg, Zn, B, Al, In, Si

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1 – 43

**Chiral *N*-*tert*-Butylsulfinyl Imines:
New Discoveries**

Starting from an enantiomer of the imine
the addition of an organometallic reagent

leads mainly to one diastereomer and
after final deprotection to an enantiomer
